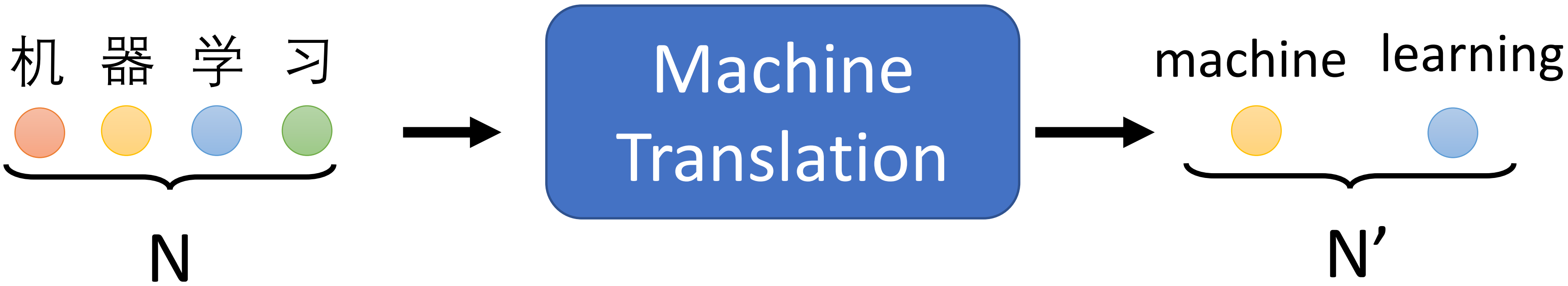
Transformer

Sequence-to-sequence (Seq2seq)

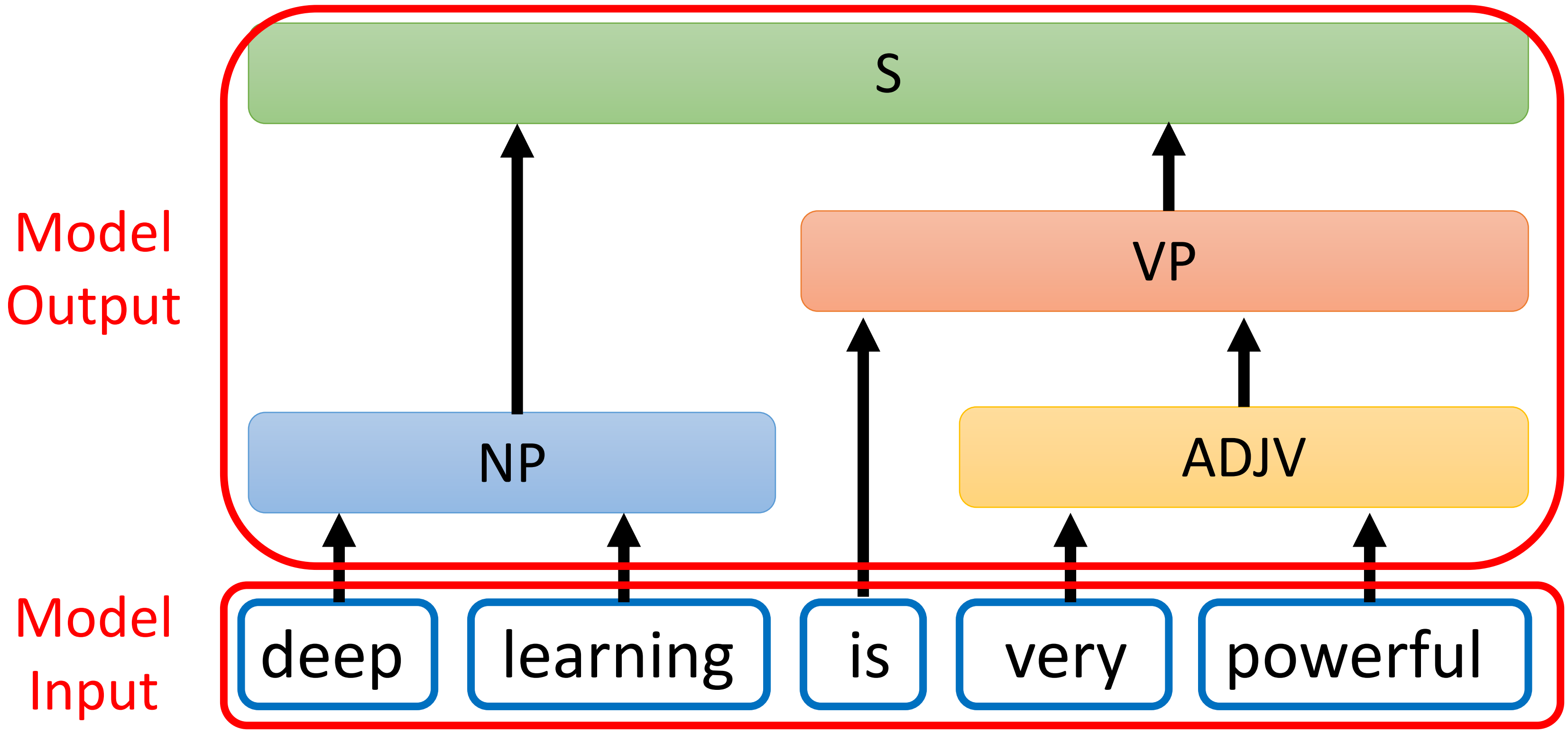
Input a sequence, output a sequence

The output length is determined by model.

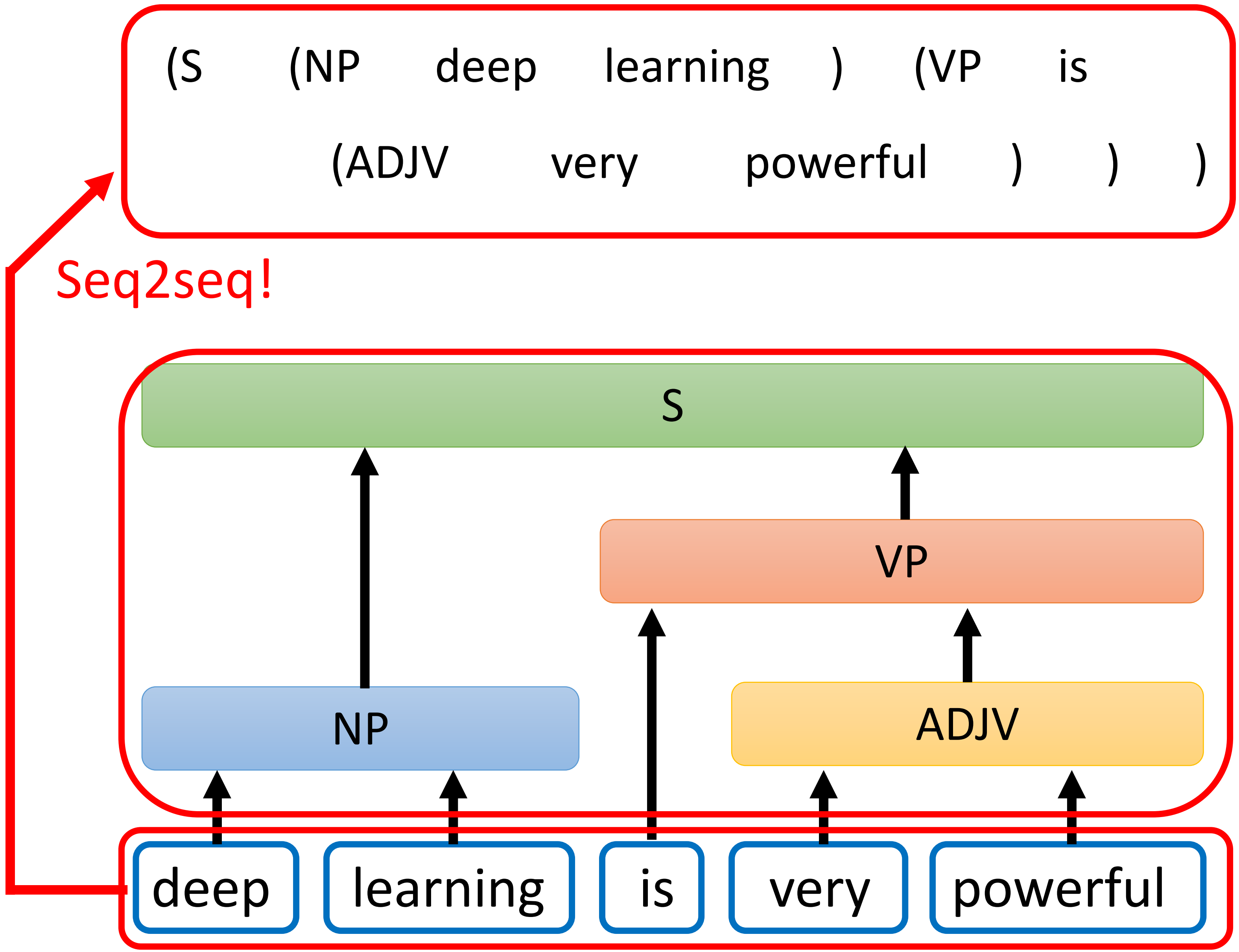


Seq2seq for Syntactic Parsing

Is it a sequence?



Seq2seq for Syntactic Parsing



Seq2seq for Syntactic Parsing

(S (NP deep learning) (VP is (ADJV very powerful)))

Grammar as a Foreign Language

Oriol Vinyals*
Google
vinyals@google.com

Lukasz Kaiser*
Google
lukaszkaiser@google.com

Terry Koo
Google
terrykoo@google.com

Slav Petrov
Google
slav@google.com

Ilya Sutskever
Google
ilyasu@google.com

Geoffrey Hinton
Google
geoffhinton@google.com

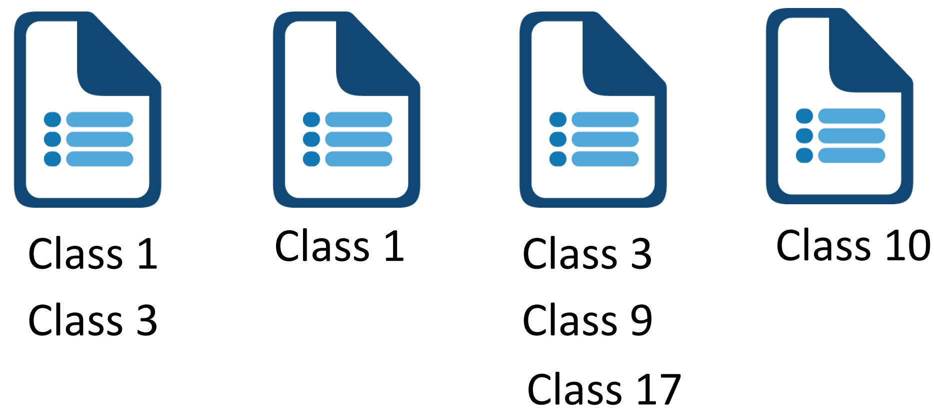
https://arxiv.org/abs/1412.7449

deep learning is very powerful

Seq2seq for Multi-label Classification

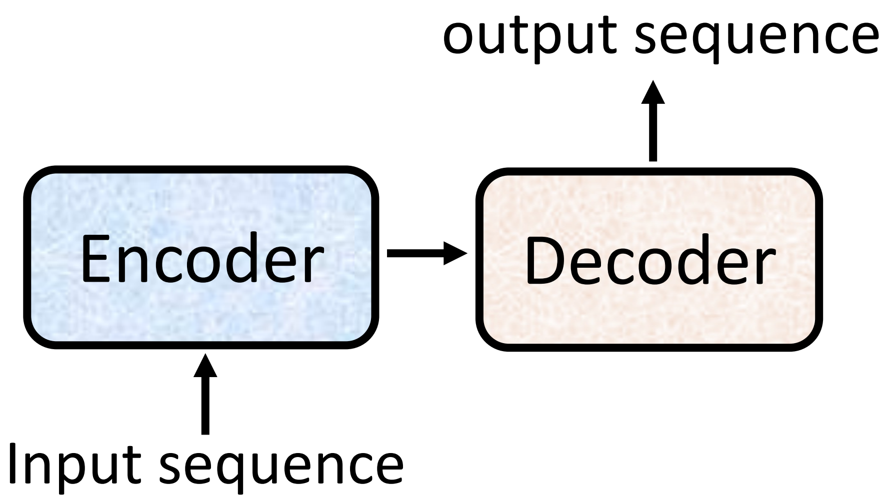
c.f. Multi-class Classification

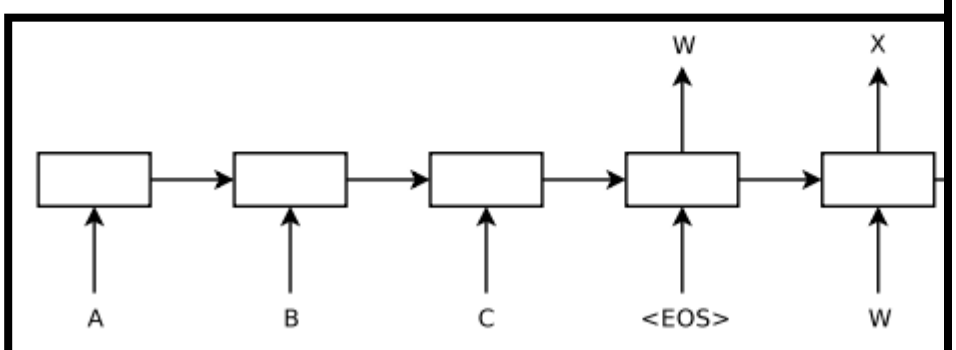
An object can belong to multiple classes.



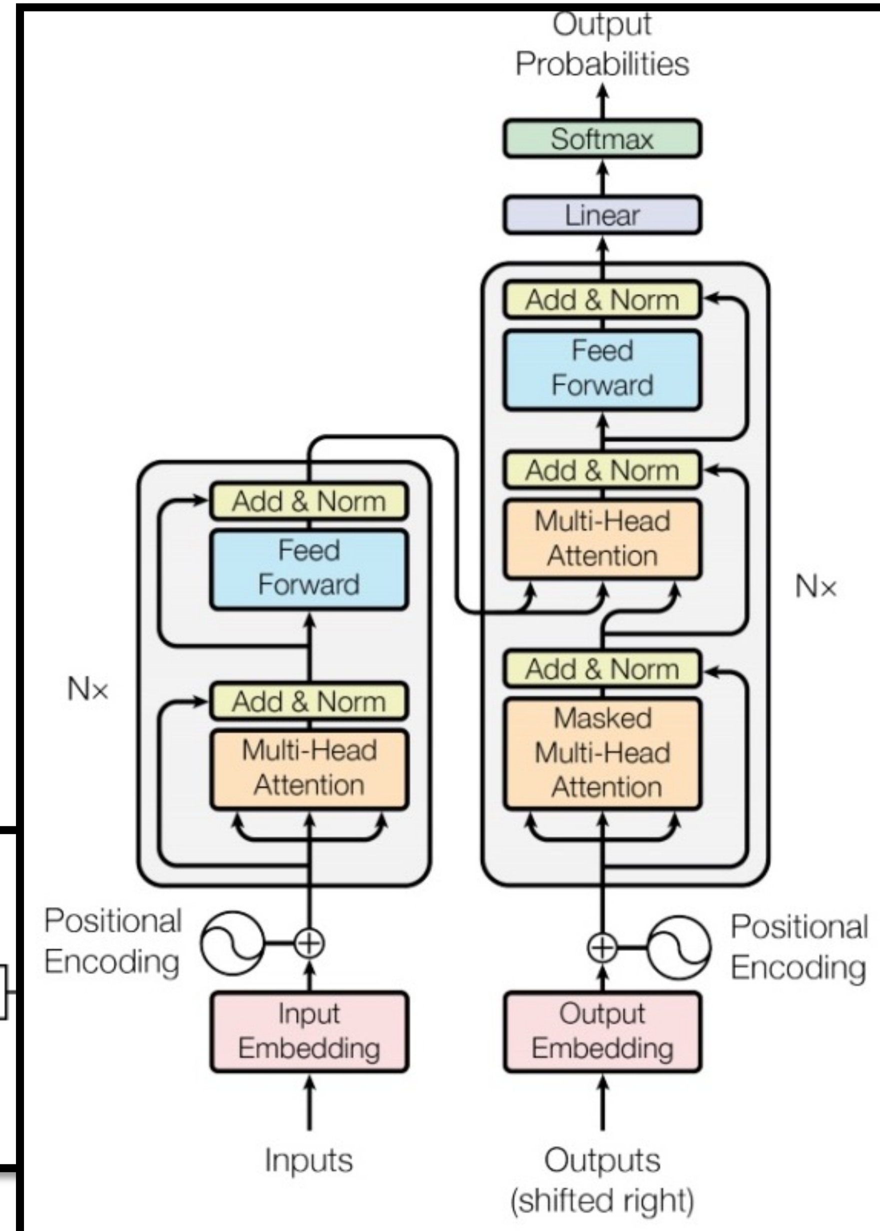
https://arxiv.org/abs/1909.03434
https://arxiv.org/abs/1707.05495

Seq2seq



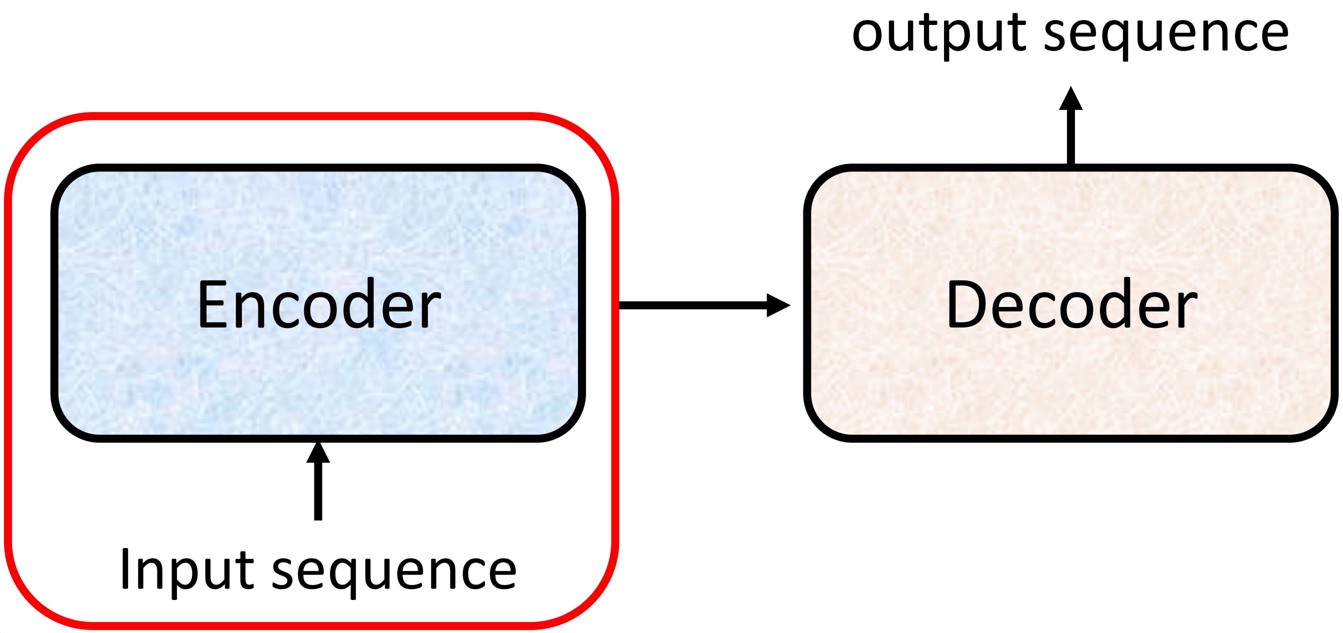


Sequence to Sequence Learning with Neural Networks
https://arxiv.org/abs/1409.3215



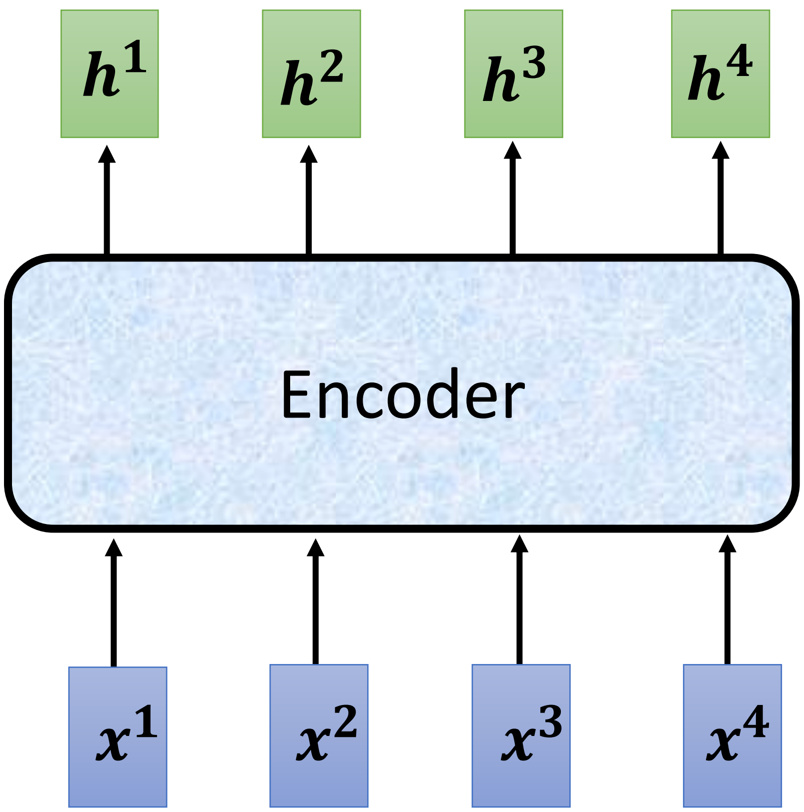
Transformer
https://arxiv.org/abs/1706.03762

Encoder

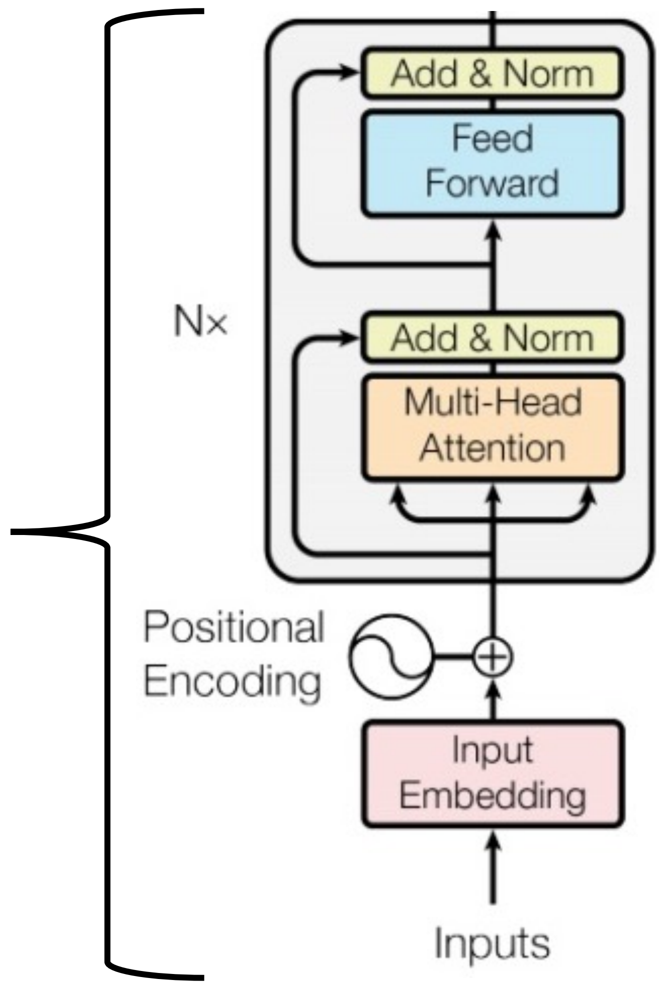


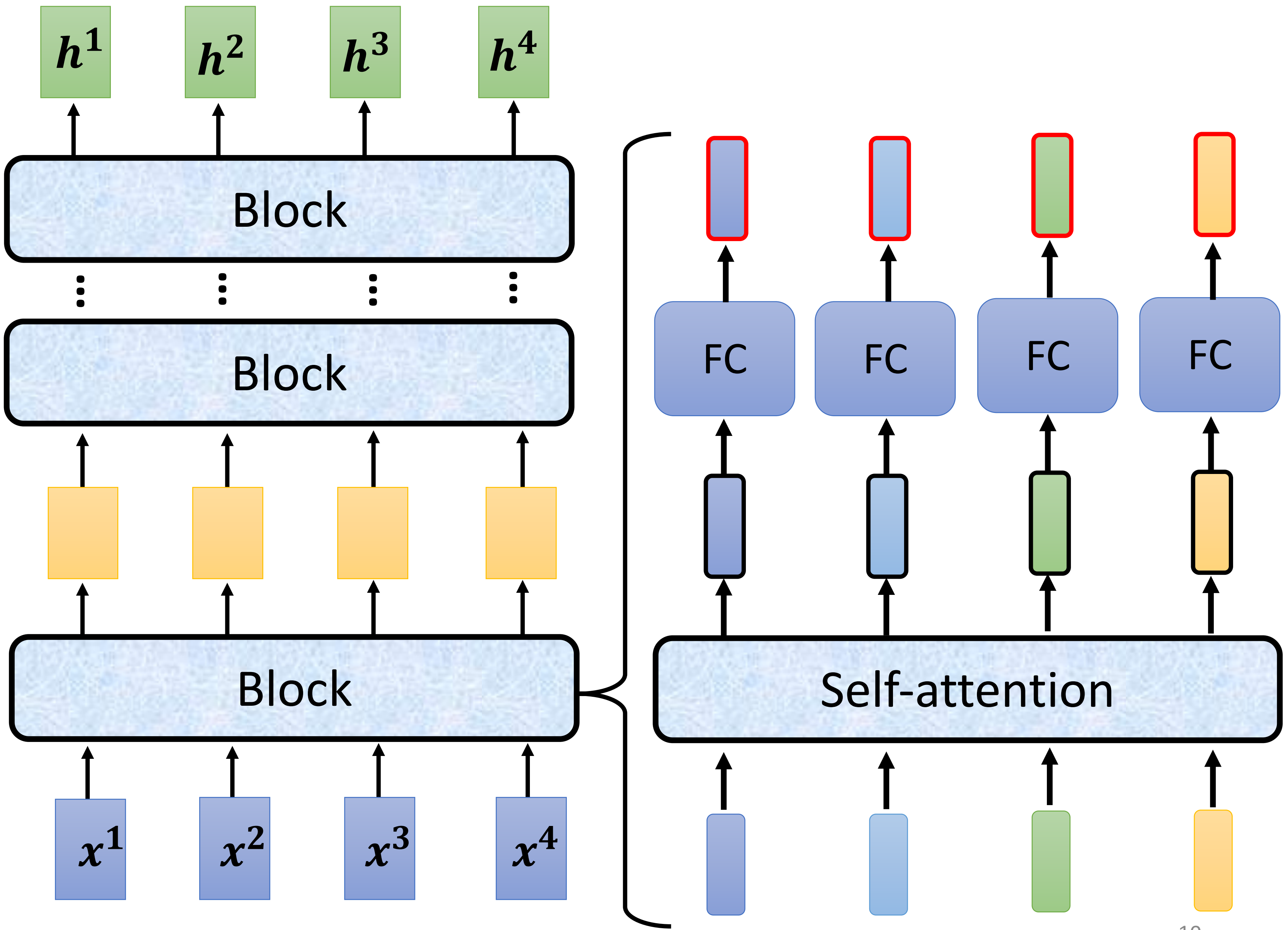
Encoder

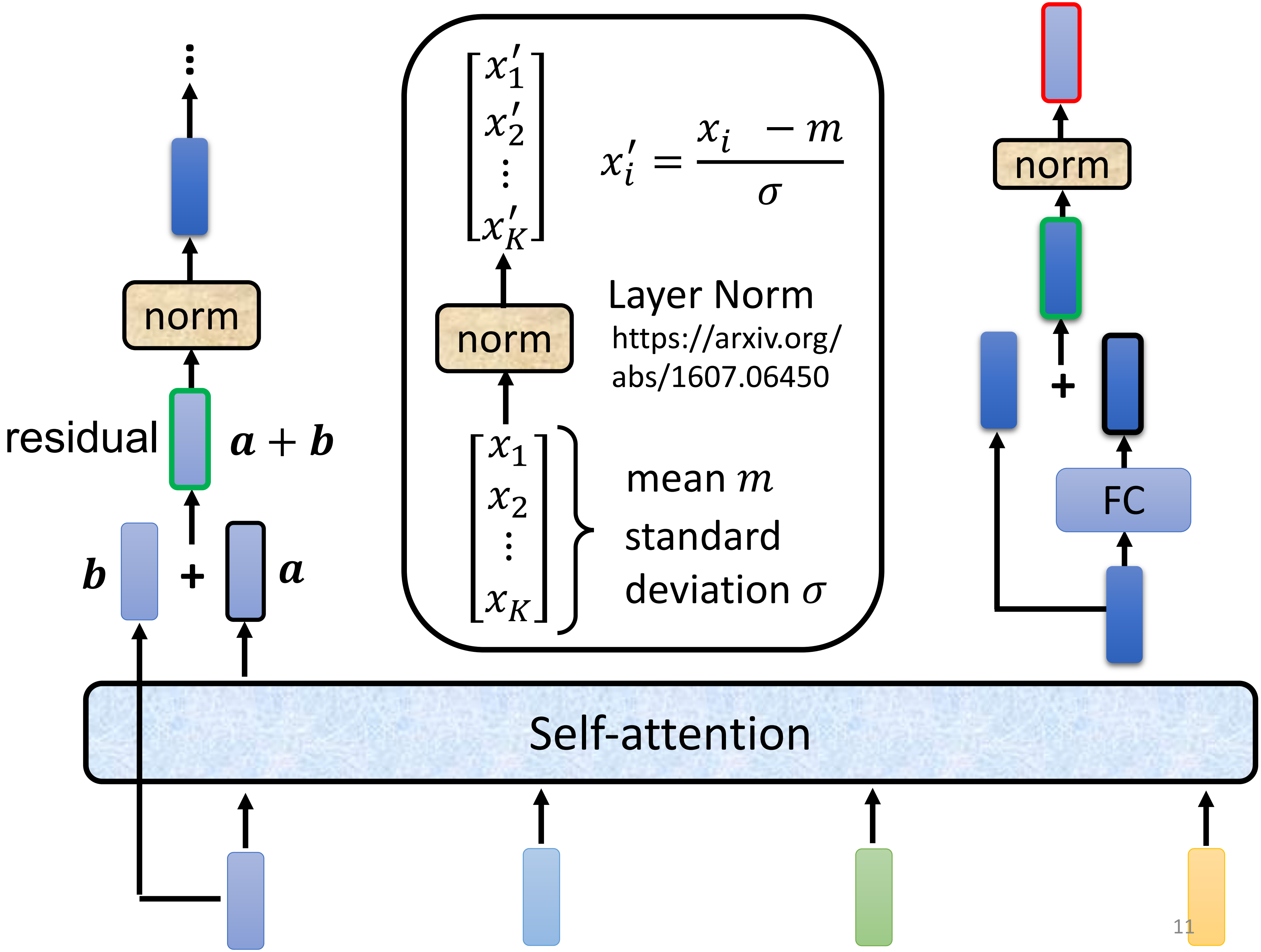
You can use **RNN** or **CNN**.



Transformer’s Encoder

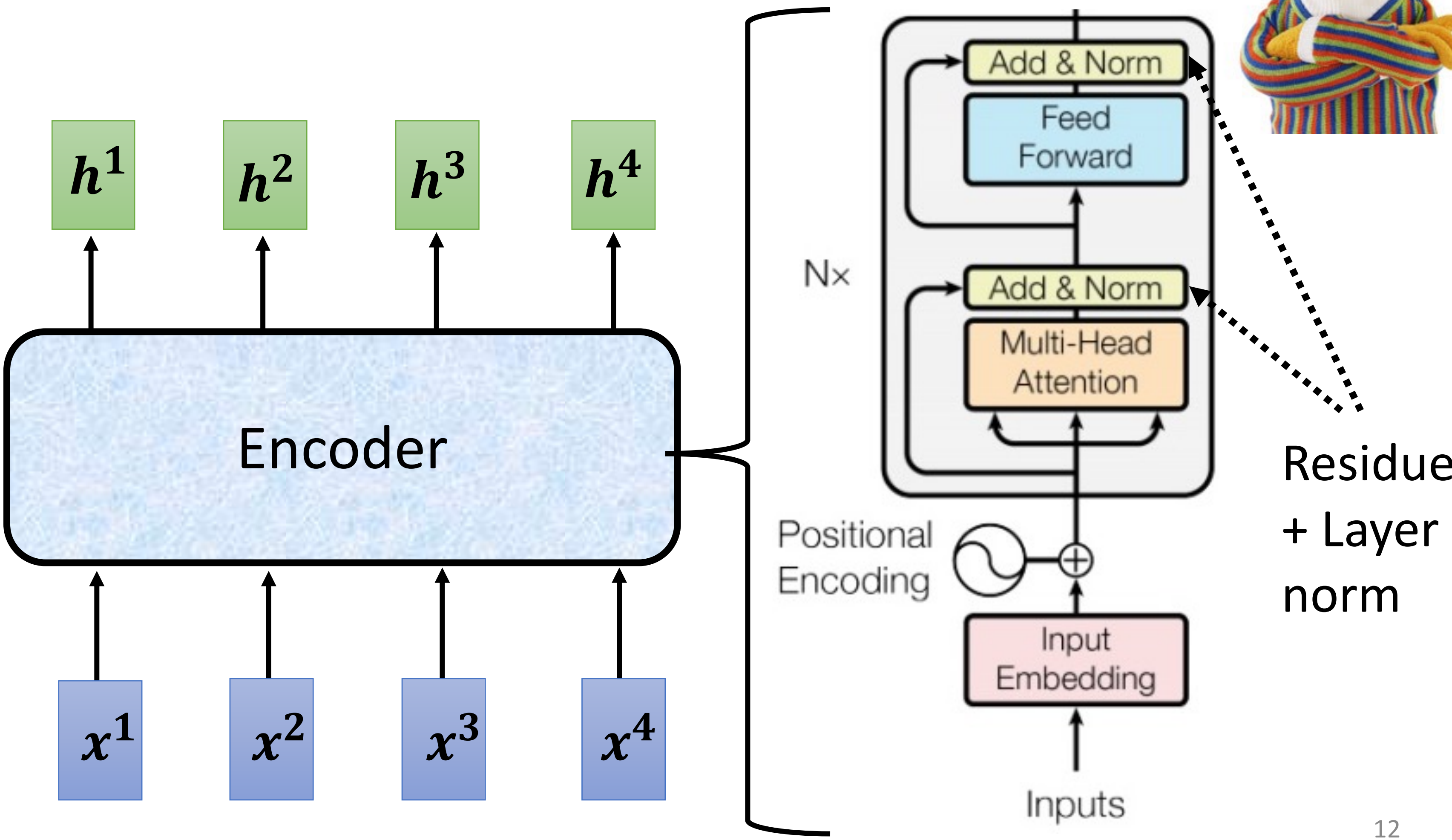



h^1
h^2
h^3
h^4
Block
Block
Block
FC
FC
FC
FC
Self-attention
x^1
x^2
x^3
x^4


norm
residual
$a + b$
b
+
a
$\begin{bmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_K \end{bmatrix}$
$x'_i = \dfrac{x_i - m}{\sigma}$
norm
Layer Norm
https://arxiv.org/abs/1607.06450
$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_K \end{bmatrix}$
mean m
standard deviation σ
norm
+
FC
Self-attention

BERT

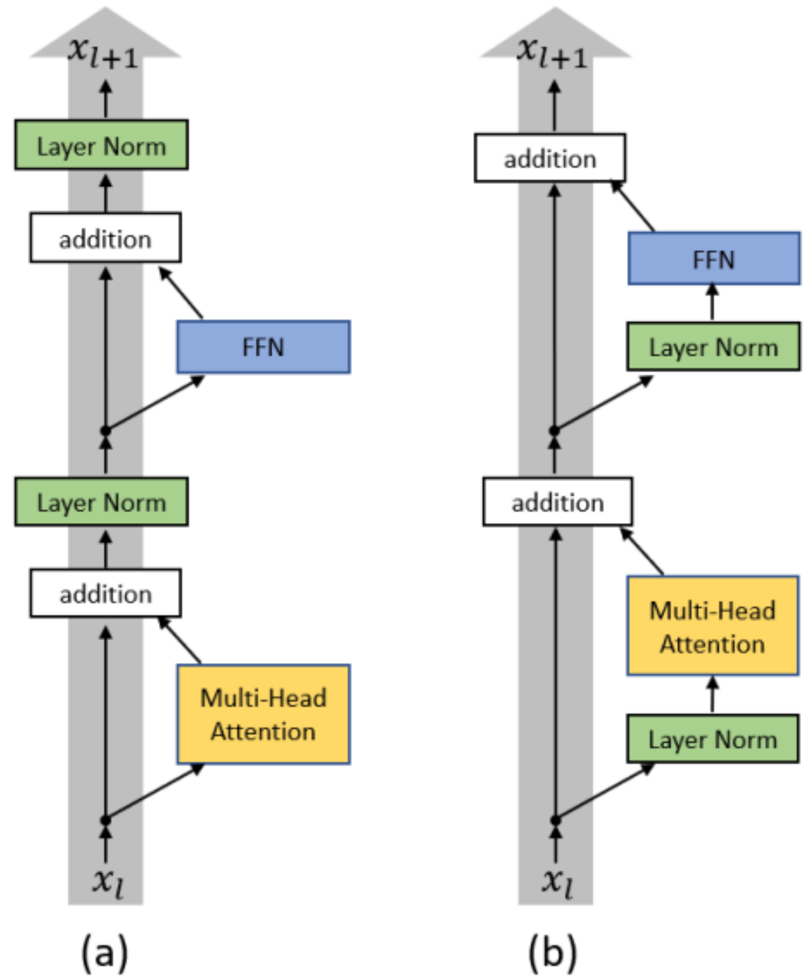
I use the **same** network architecture as **transformer encoder**.



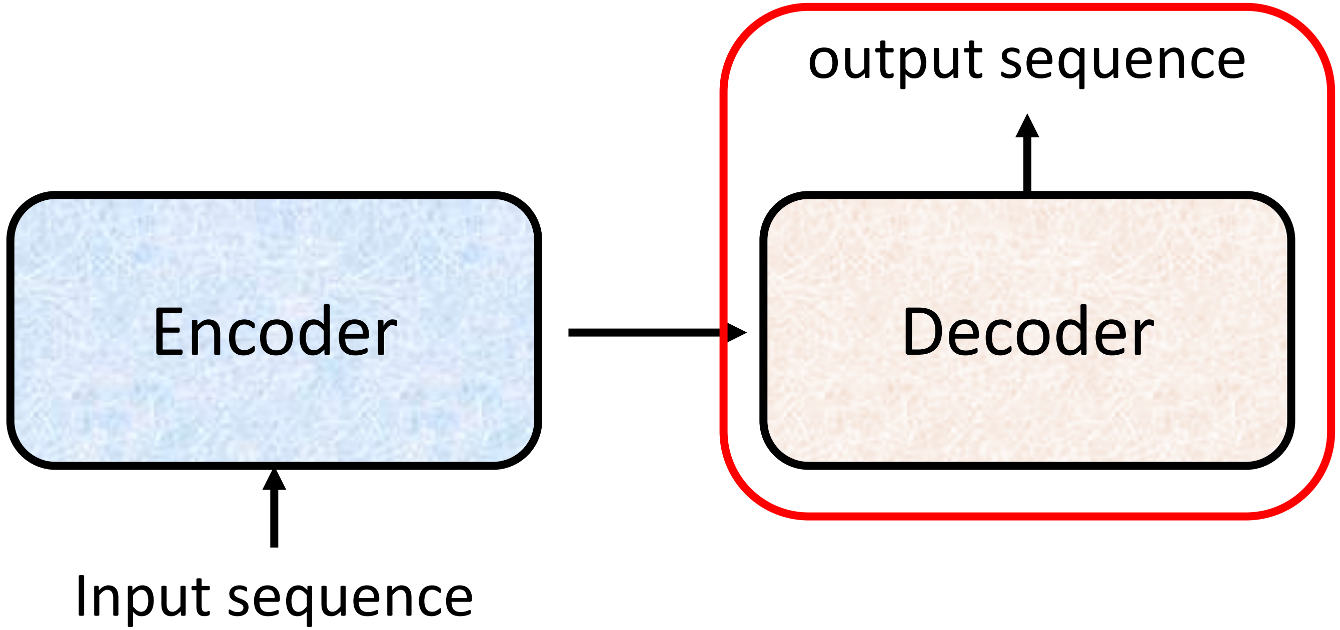
To learn more ……

- On Layer Normalization in the Transformer Architecture
- https://arxiv.org/abs/2002.04745

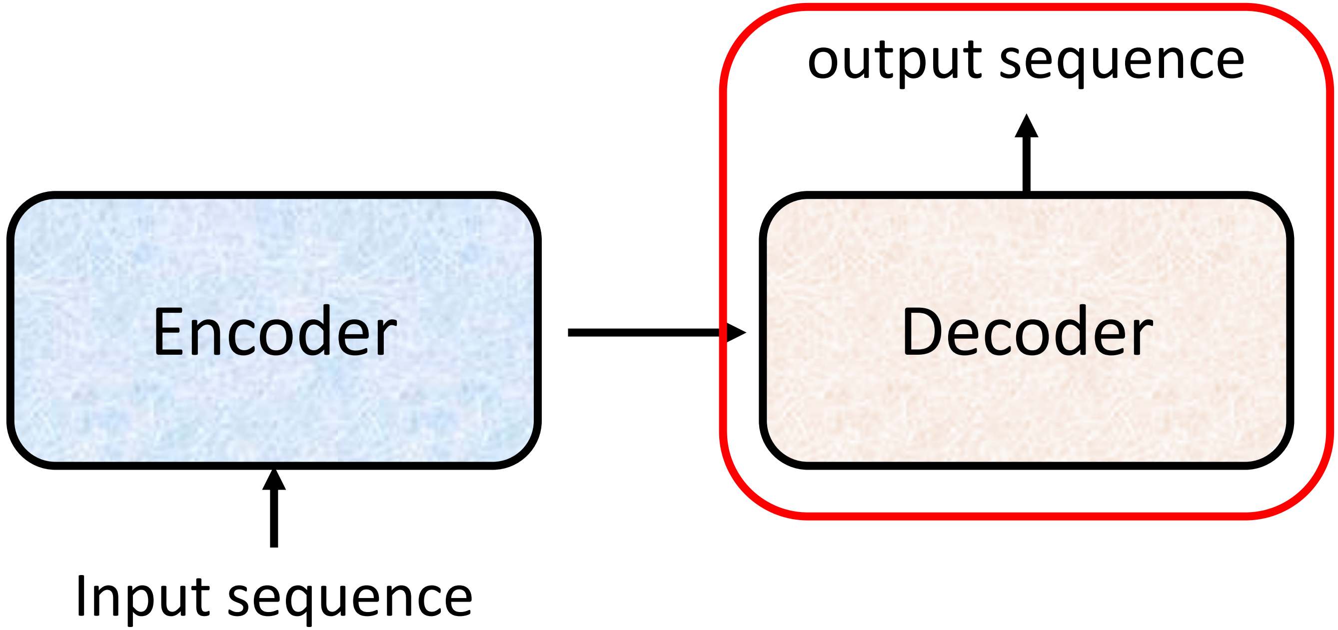
- PowerNorm: Rethinking Batch Normalization in Transformers
- https://arxiv.org/abs/2003.07845



Decoder

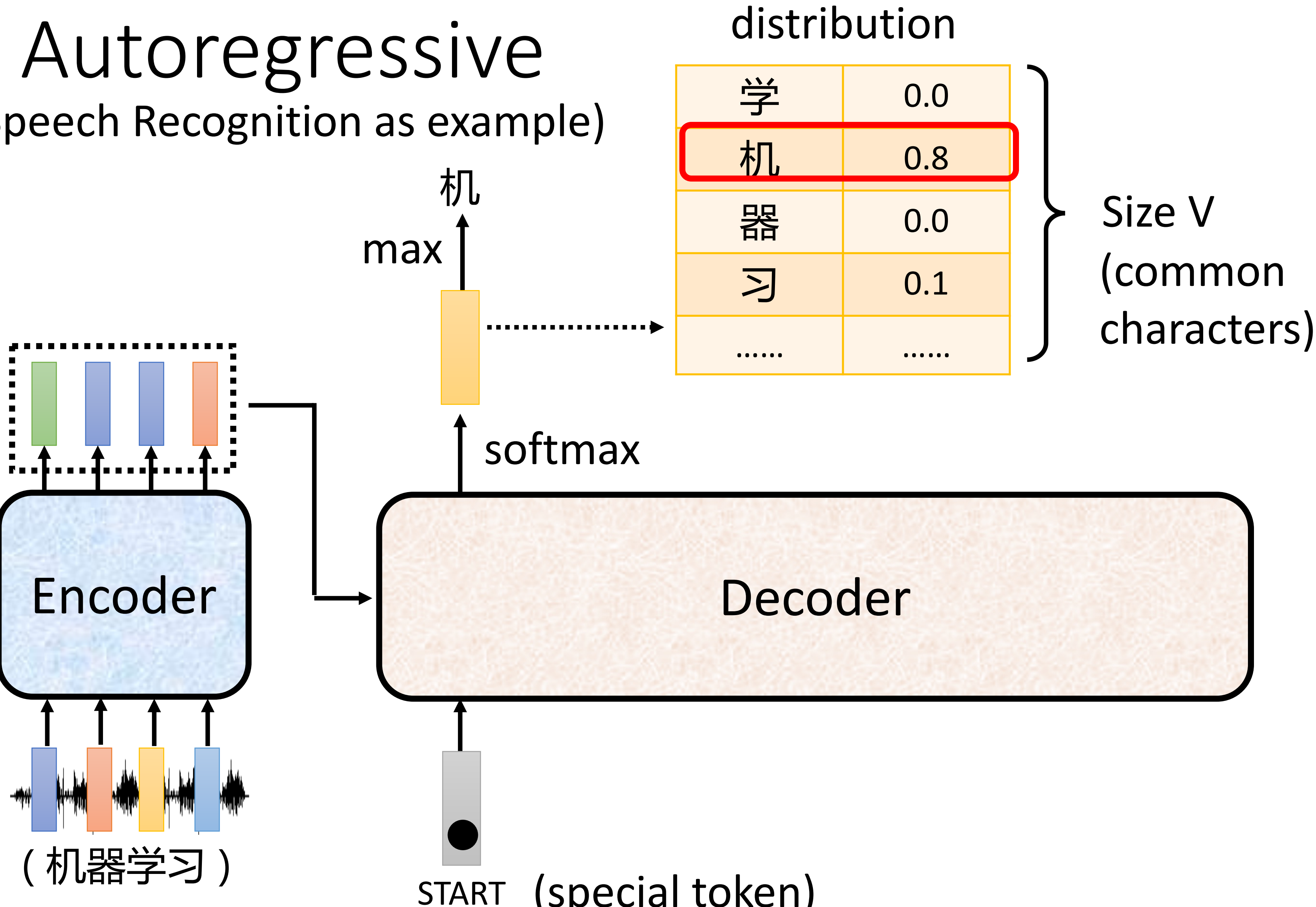


Decoder – Autoregressive (AT)

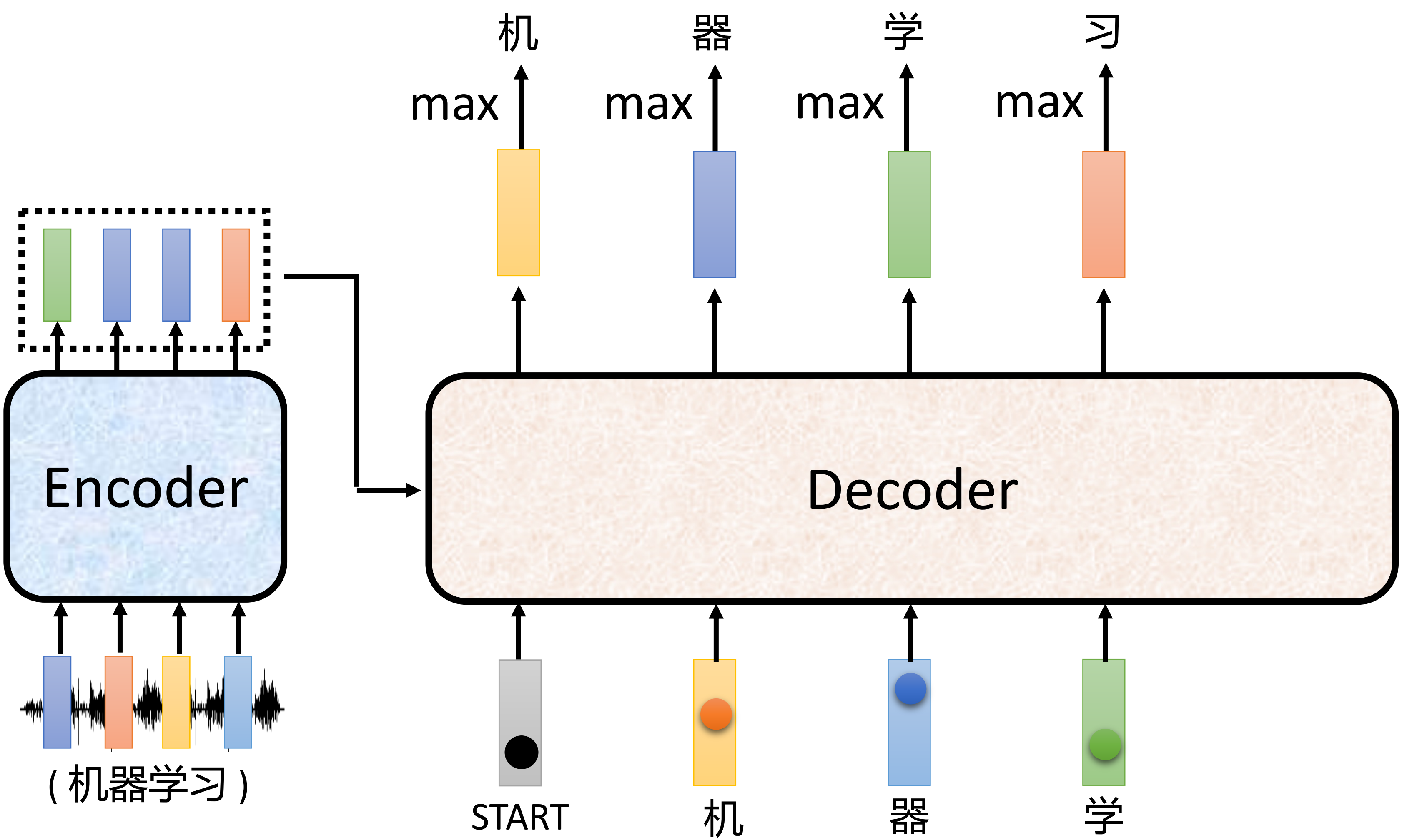


Autoregressive

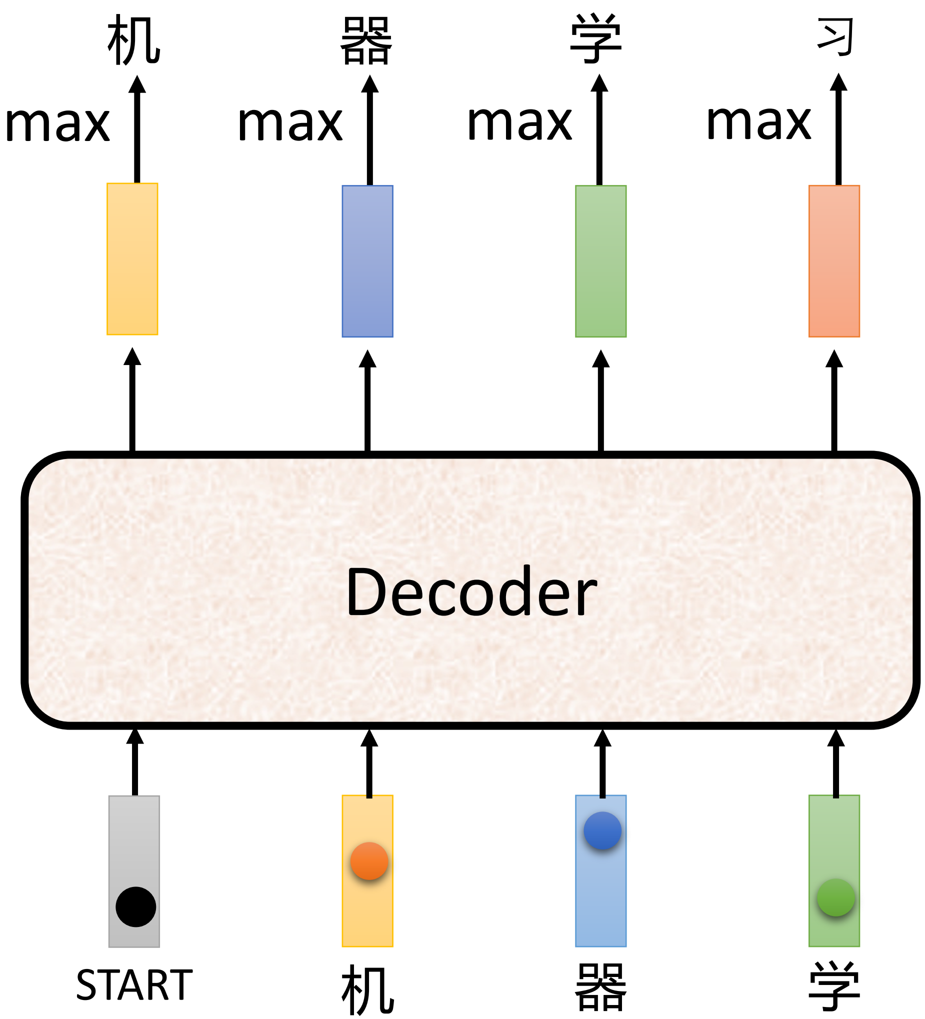
(Speech Recognition as example)

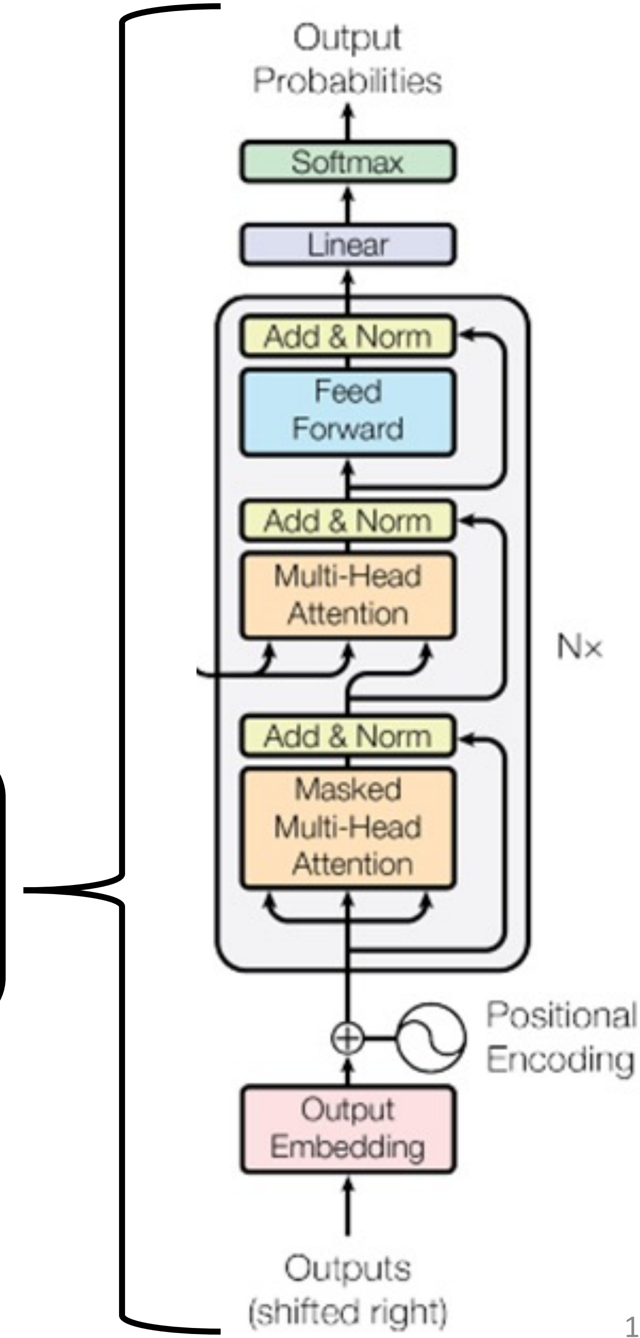


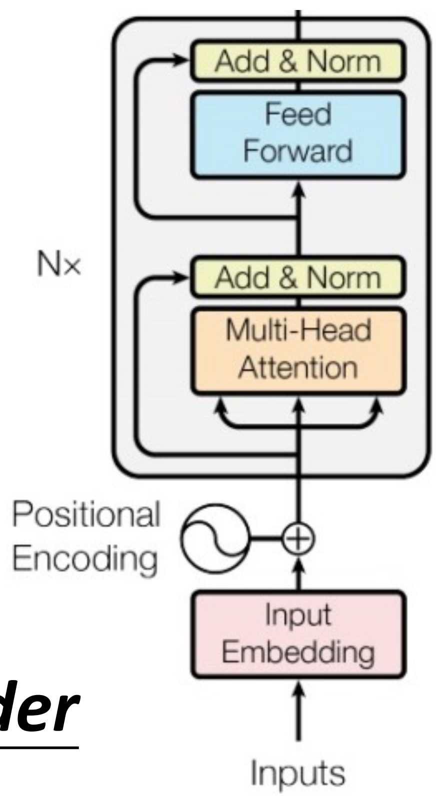
Autoregressive



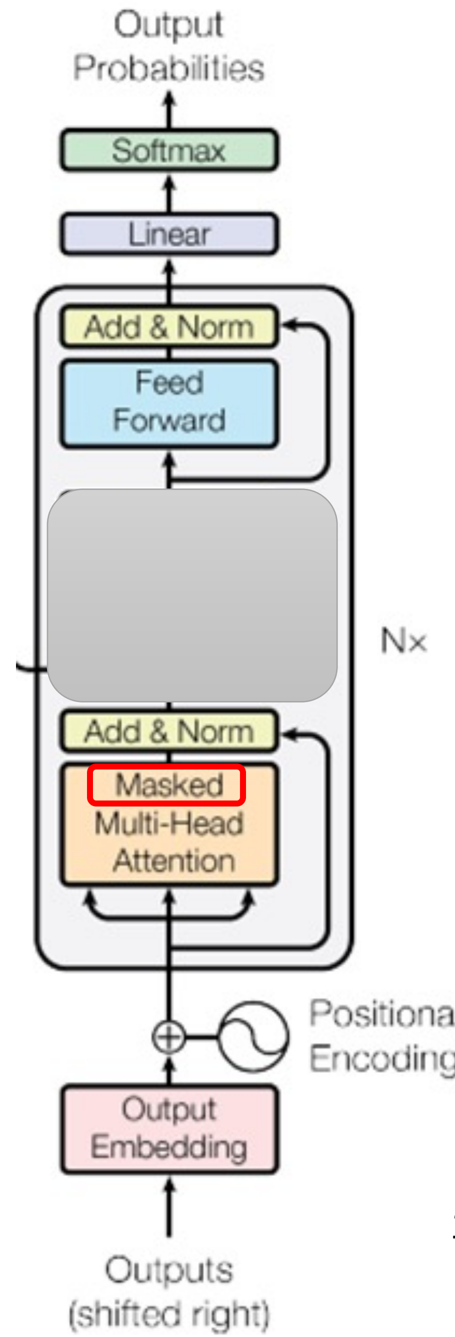
ignore the input from the encoder here ☺





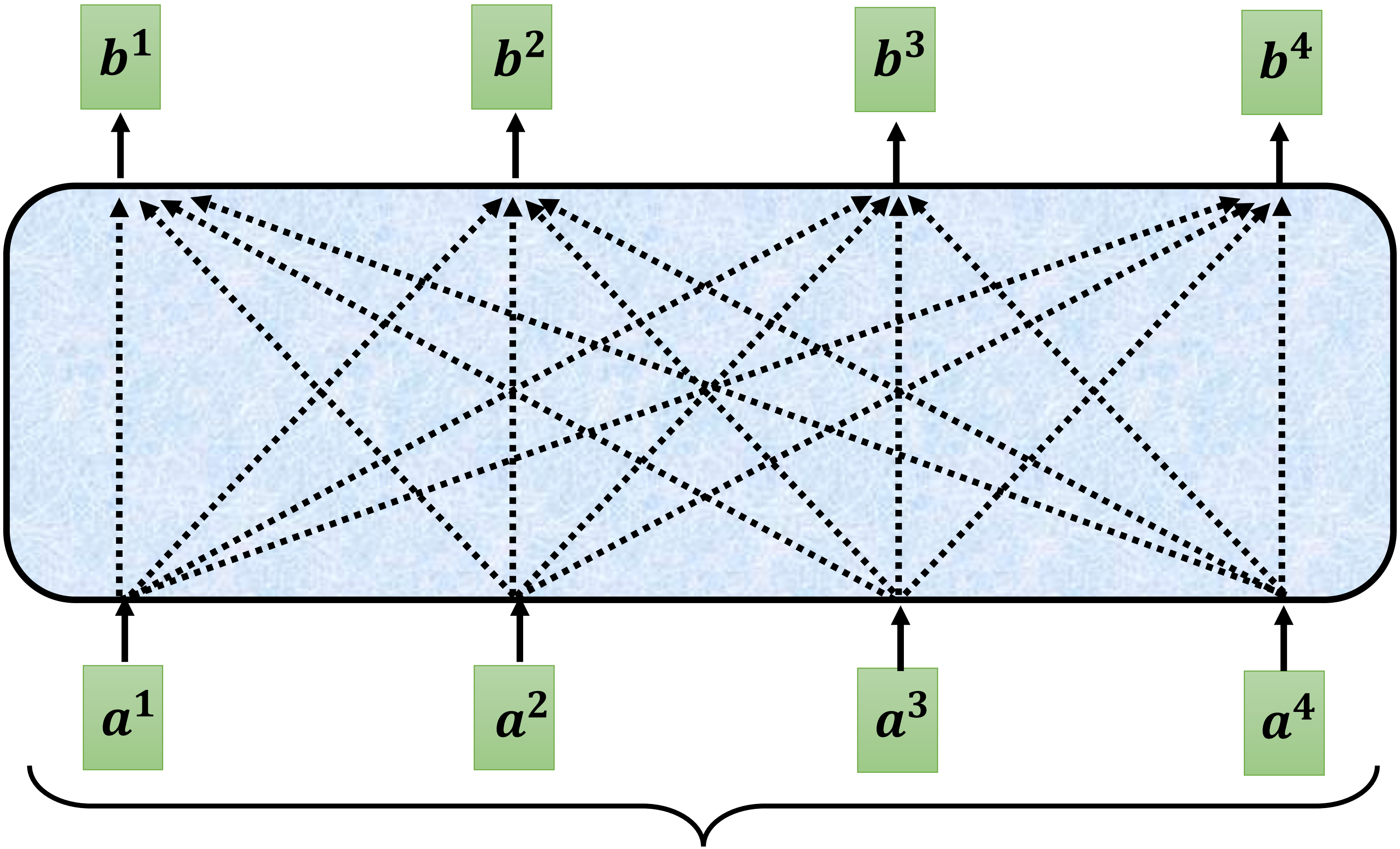


Encoder



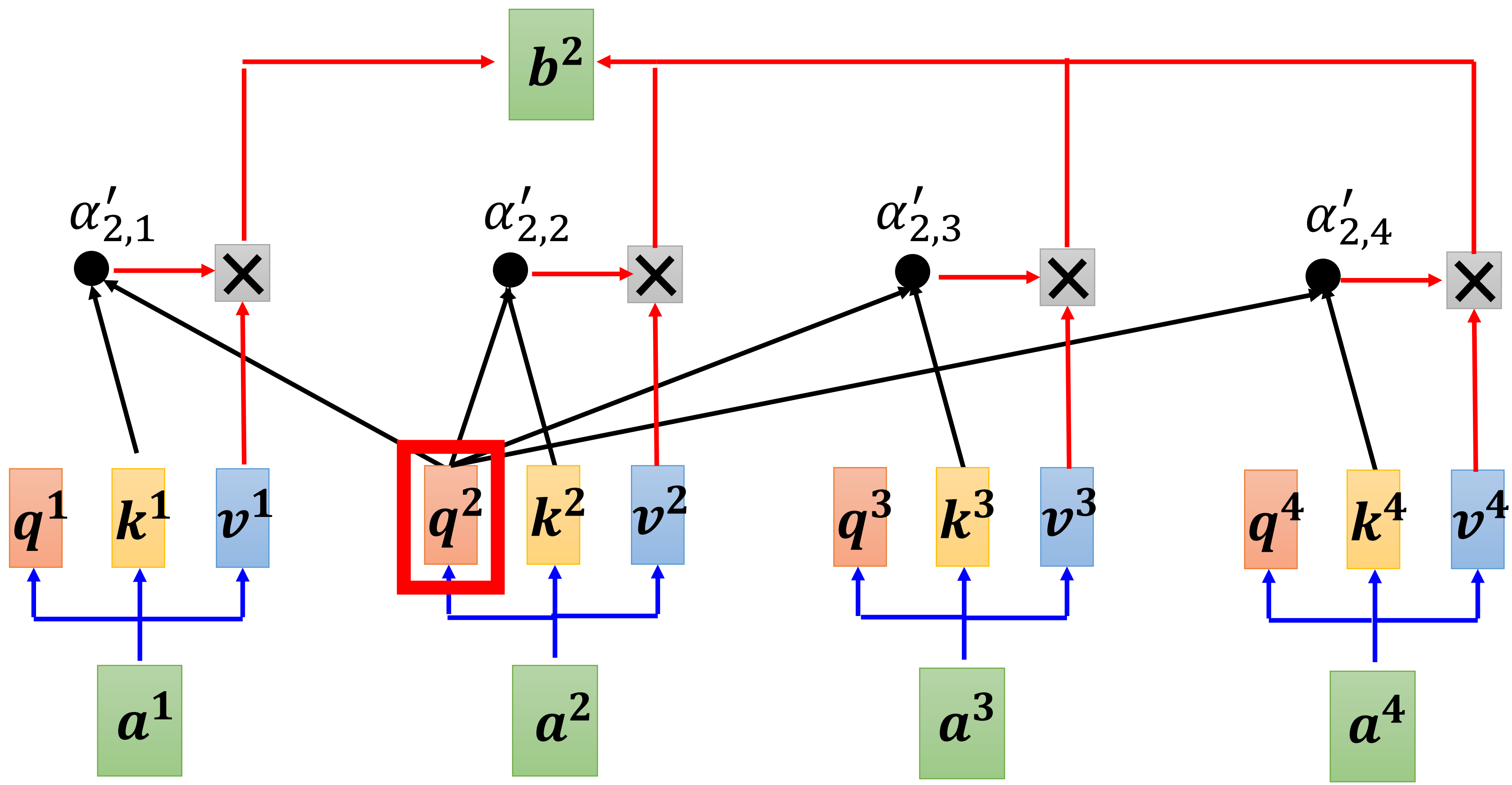
Decoder

Self-attention → Masked Self-attention



Can be either **input** or **a hidden layer**

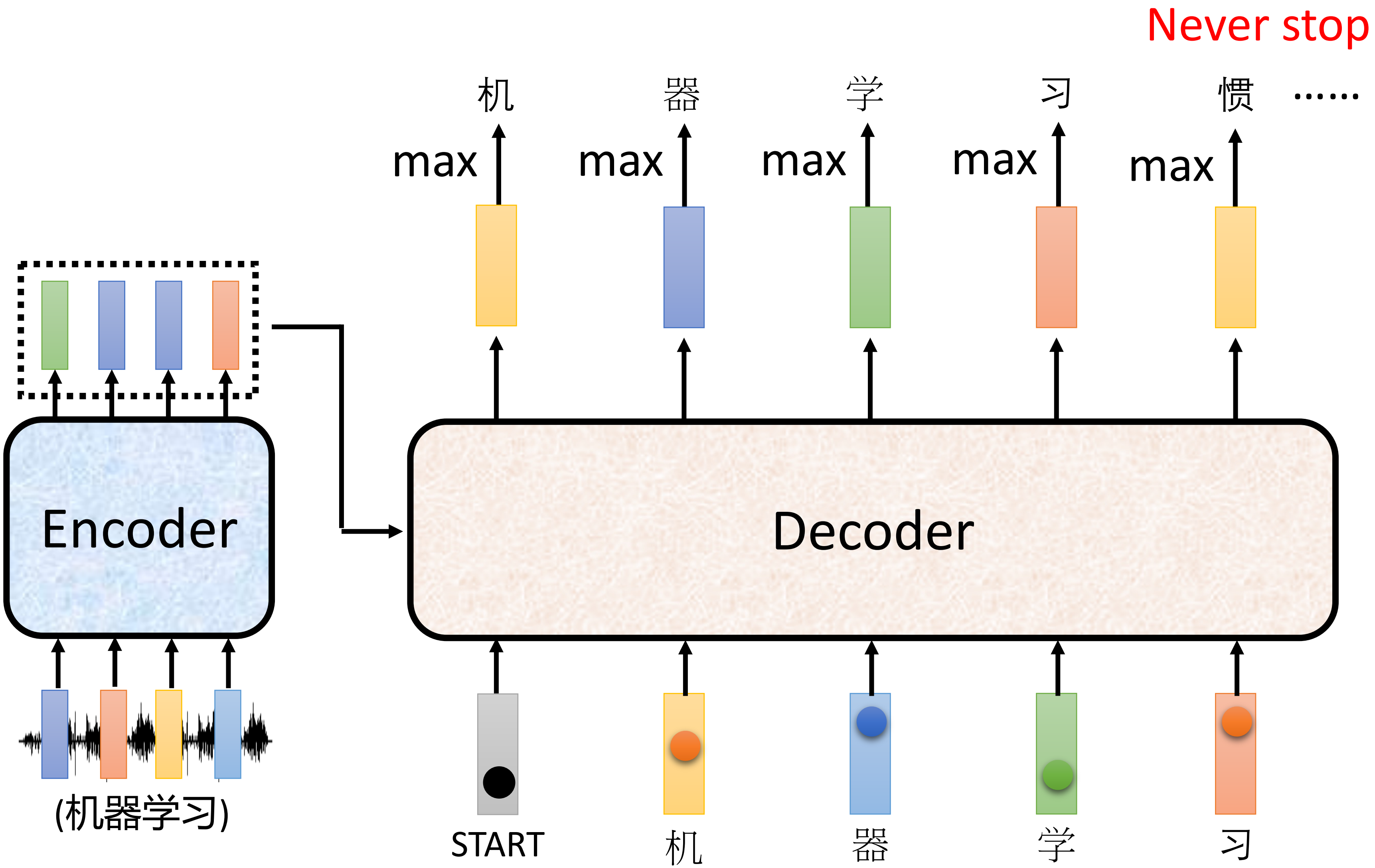
Self-attention → *Masked Self-attention*



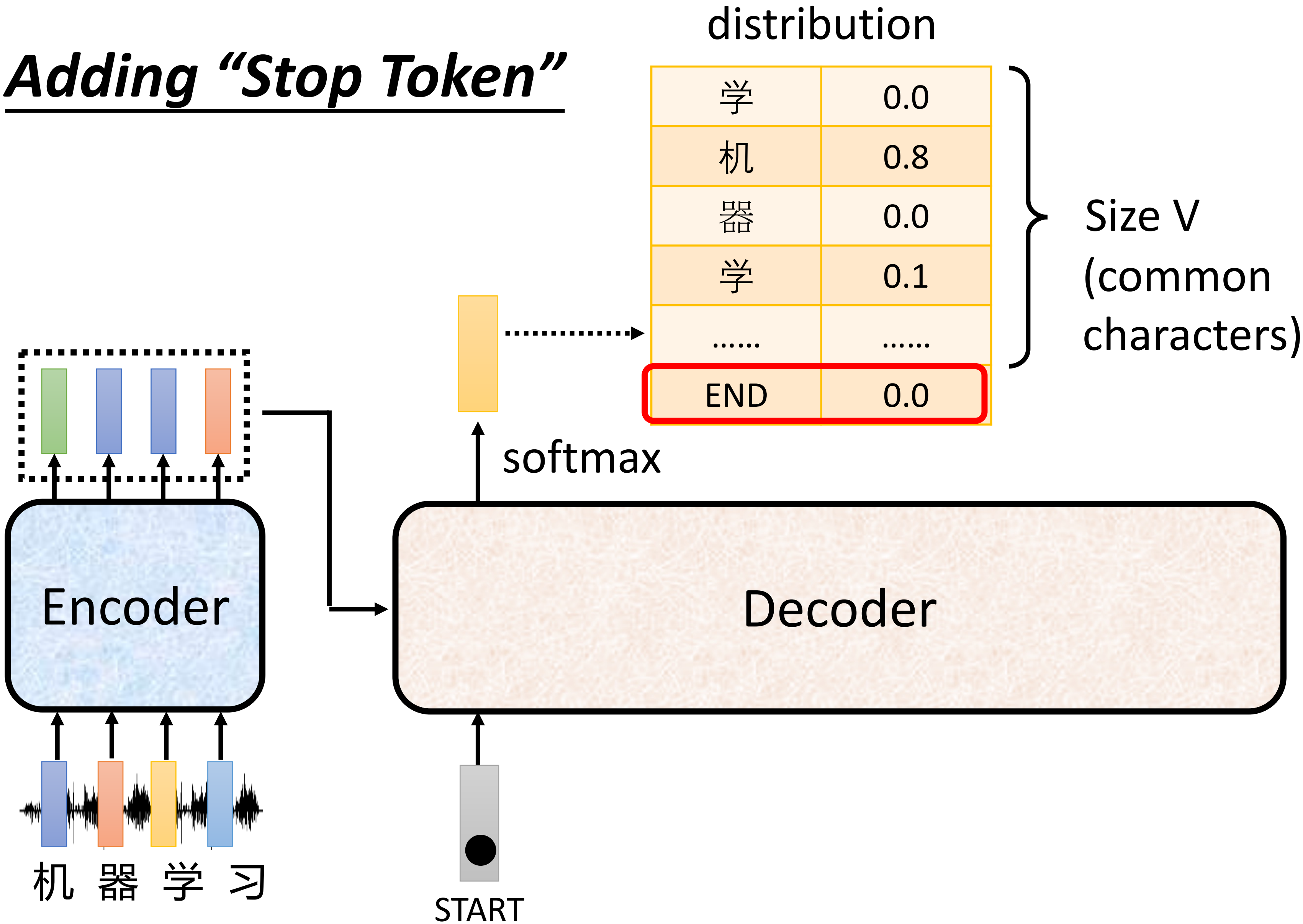
Why masked? Consider how does decoder work

Autoregressive

We do not know the correct output length.

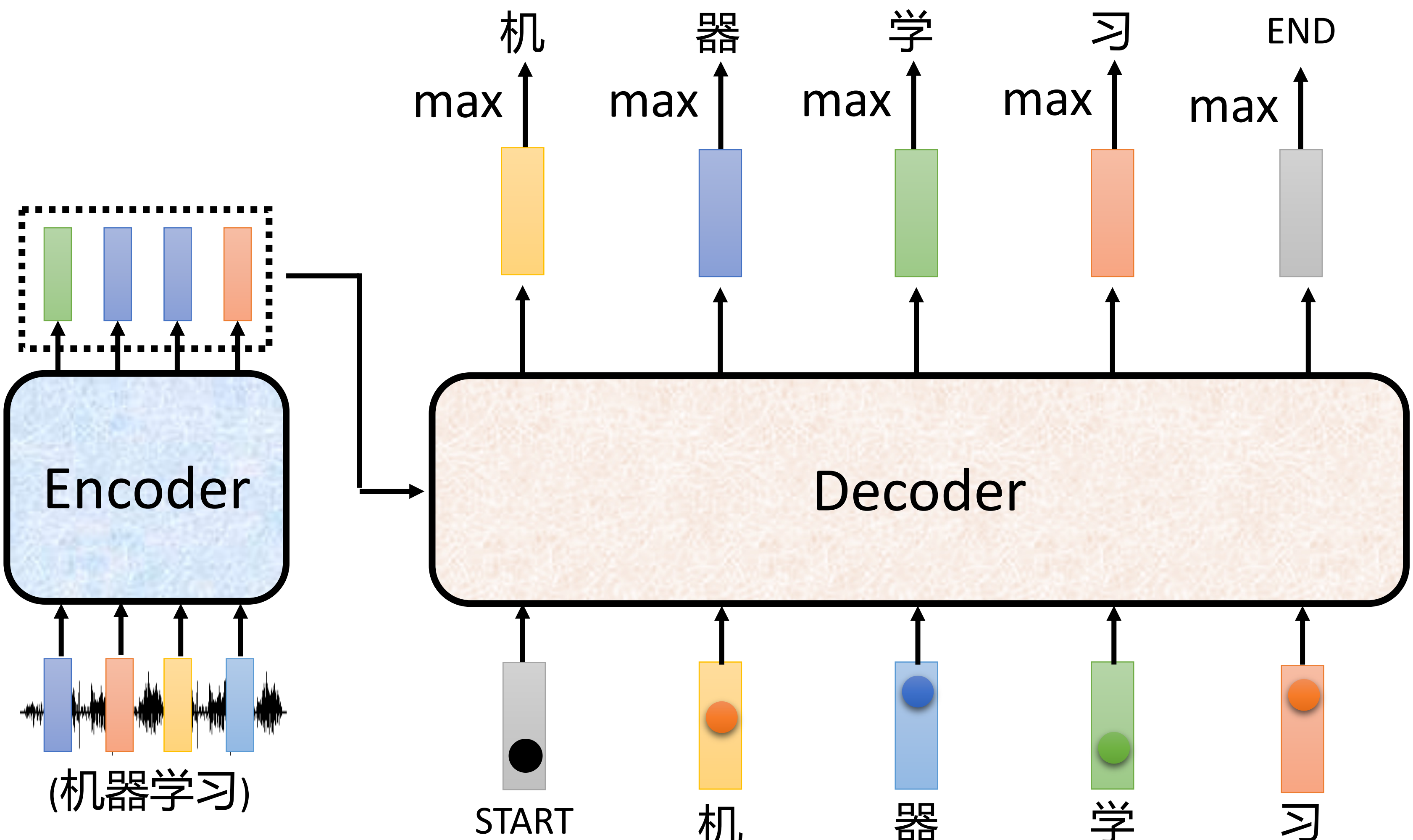


Adding "Stop Token"

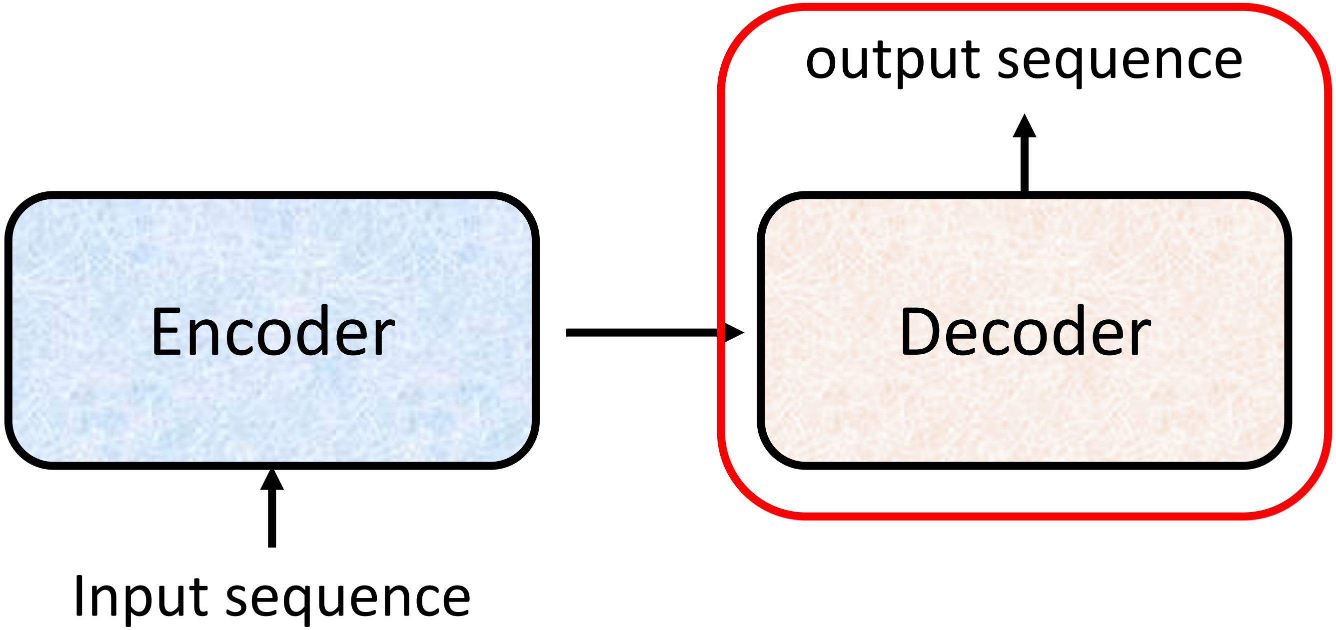


Autoregressive

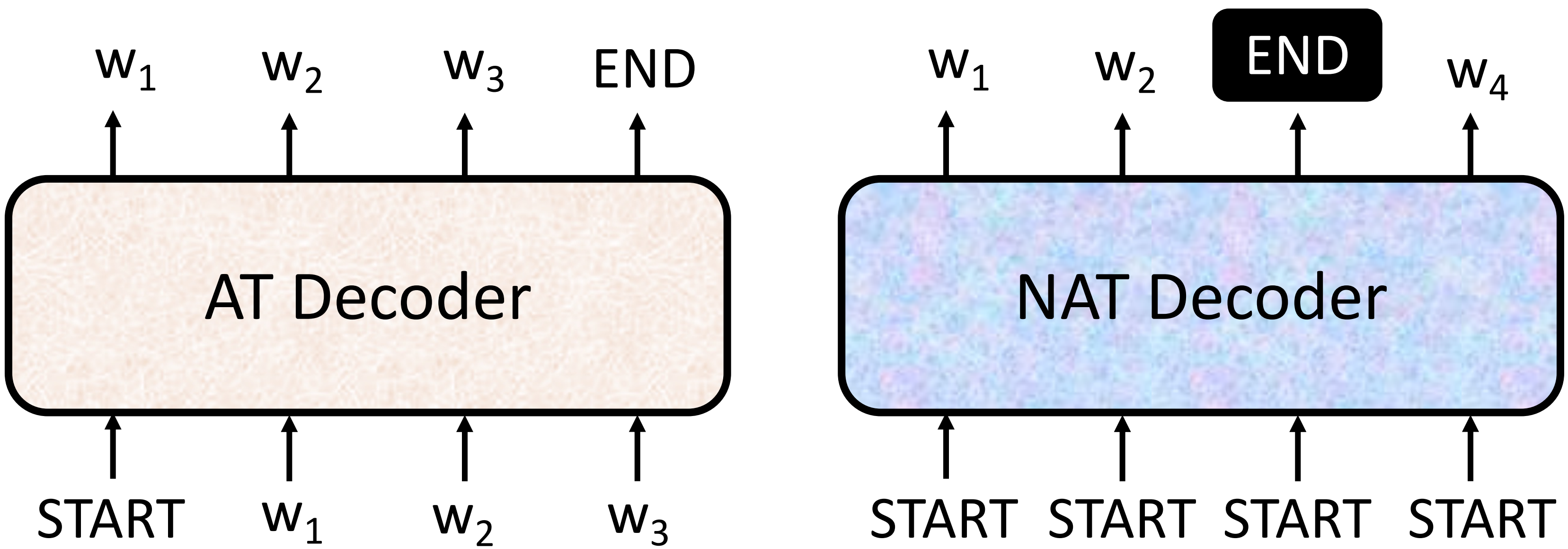
Stop at here!



Decoder
– Non-autoregressive (NAT)

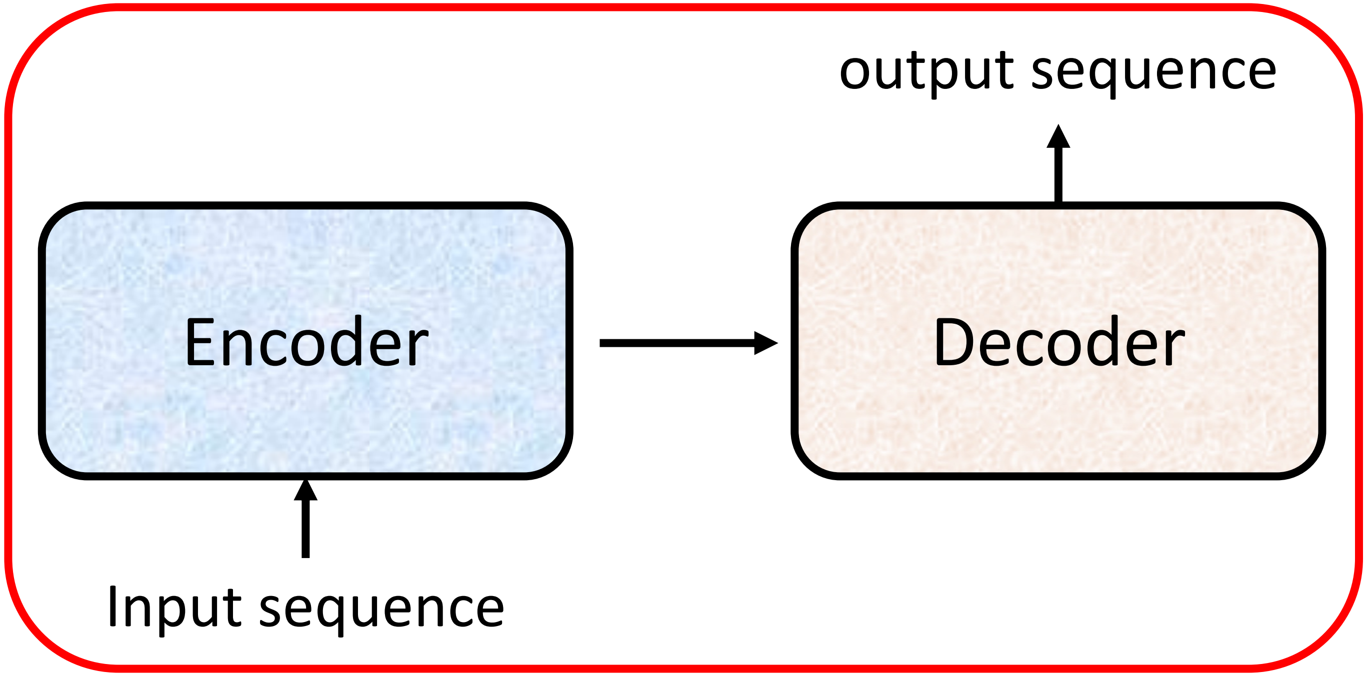


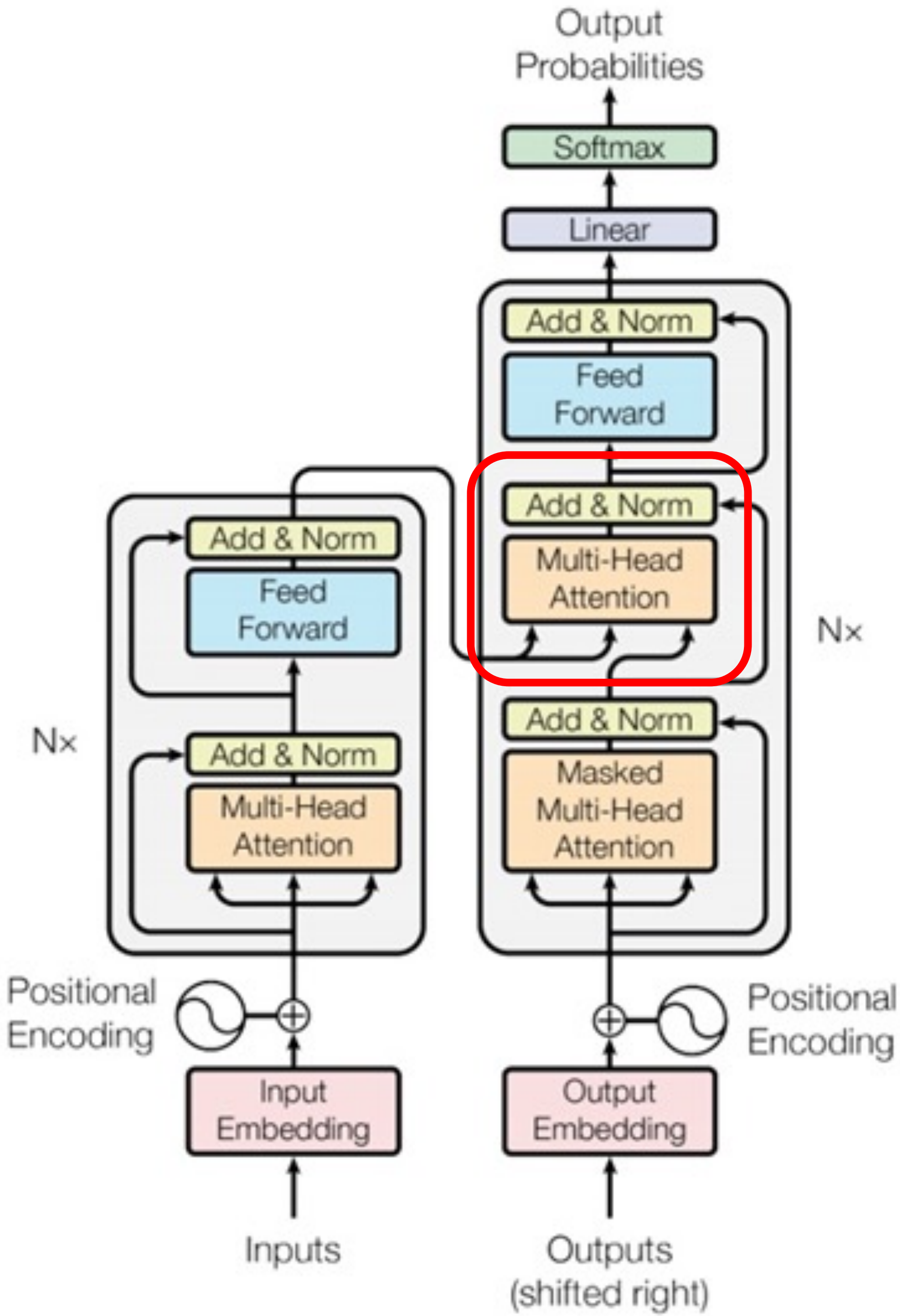
AT v.s. NAT

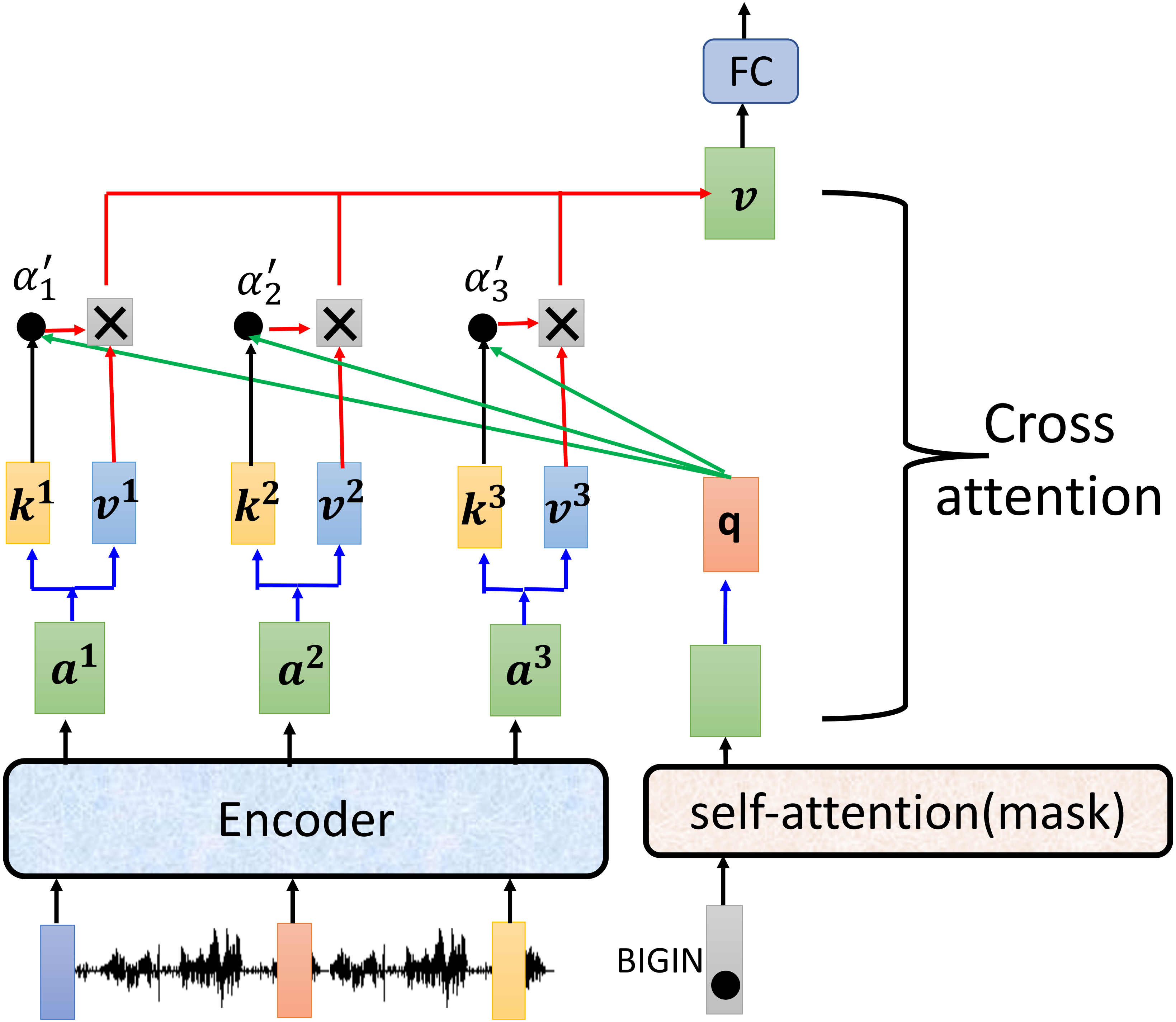


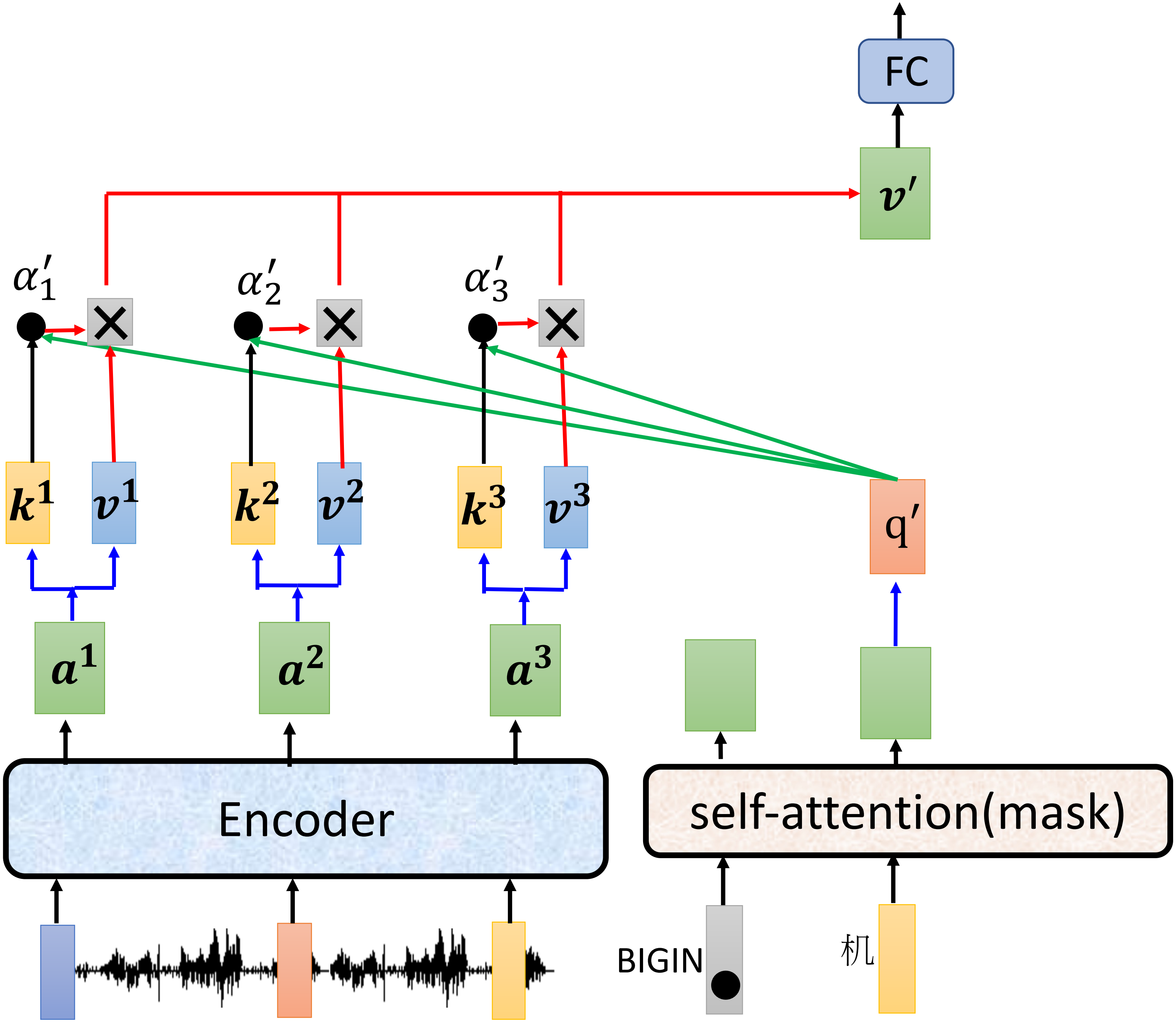
- How to decide the output length for NAT decoder?
  - Another predictor for output length
  - Output a very long sequence, ignore tokens after END
- Advantage: parallel, more stable generation (e.g., TTS)
- NAT is usually worse than AT (why? Multi-modality)

Encoder-Decoder

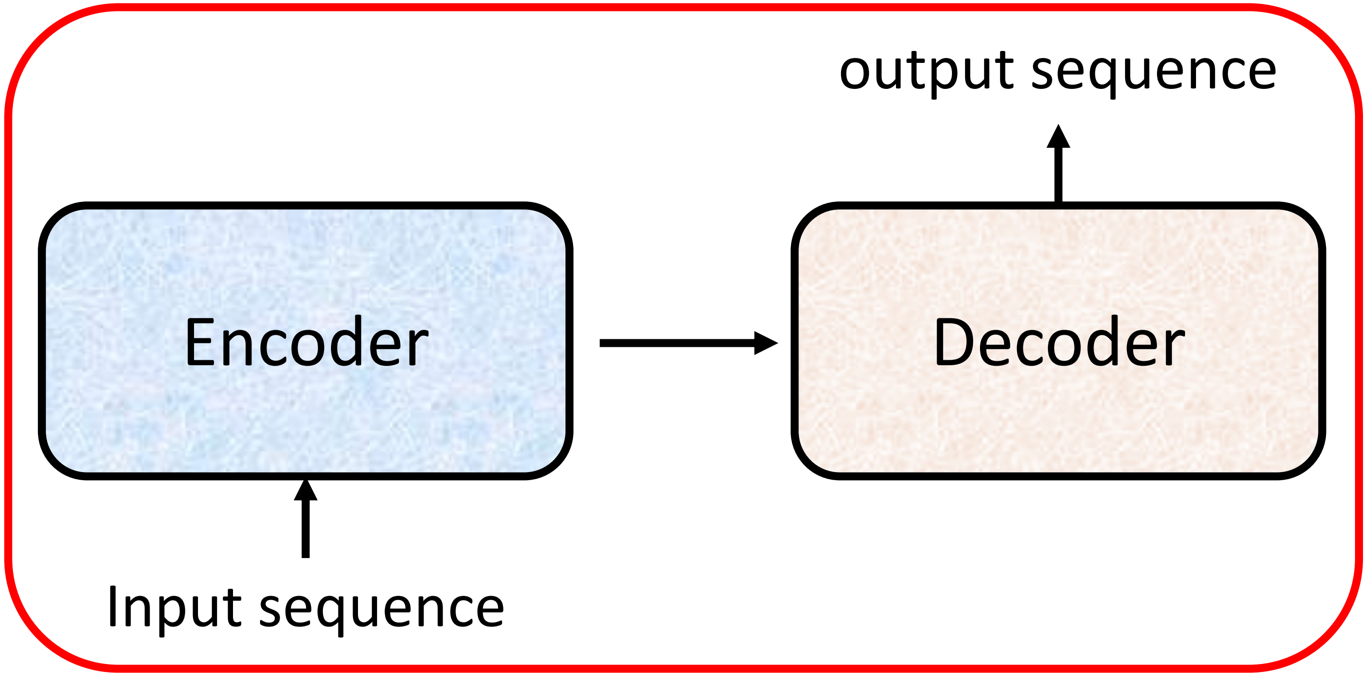



Output
Probabilities
Softmax
Linear
Add & Norm
Feed
Forward
Add & Norm
Multi-Head
Attention
Add & Norm
Masked
Multi-Head
Attention
Nx
Add & Norm
Feed
Forward
Add & Norm
Multi-Head
Attention
Nx
Positional
Encoding
Positional
Encoding
Input
Embedding
Output
Embedding
Inputs
Outputs
(shifted right)

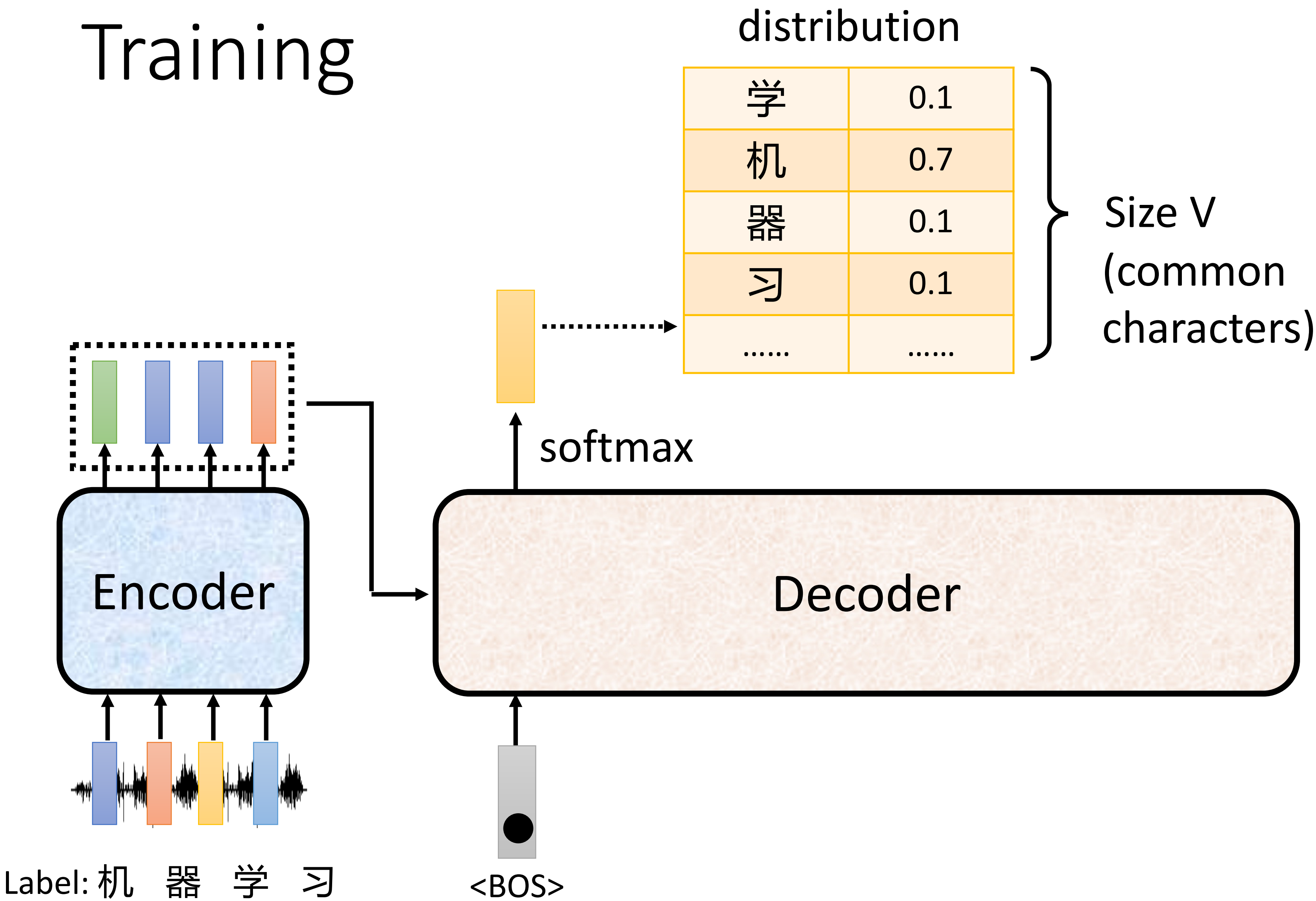

FC
v
α'_1
α'_2
α'_3
k^1 v^1
k^2 v^2
k^3 v^3
q
Cross attention
a^1
a^2
a^3
Encoder
self-attention(mask)
BIGIN


FC
v′
α′1
α′2
α′3
k1
v1
k2
v2
k3
v3
q′
a1
a2
a3
Encoder
self-attention(mask)
BIGIN
机

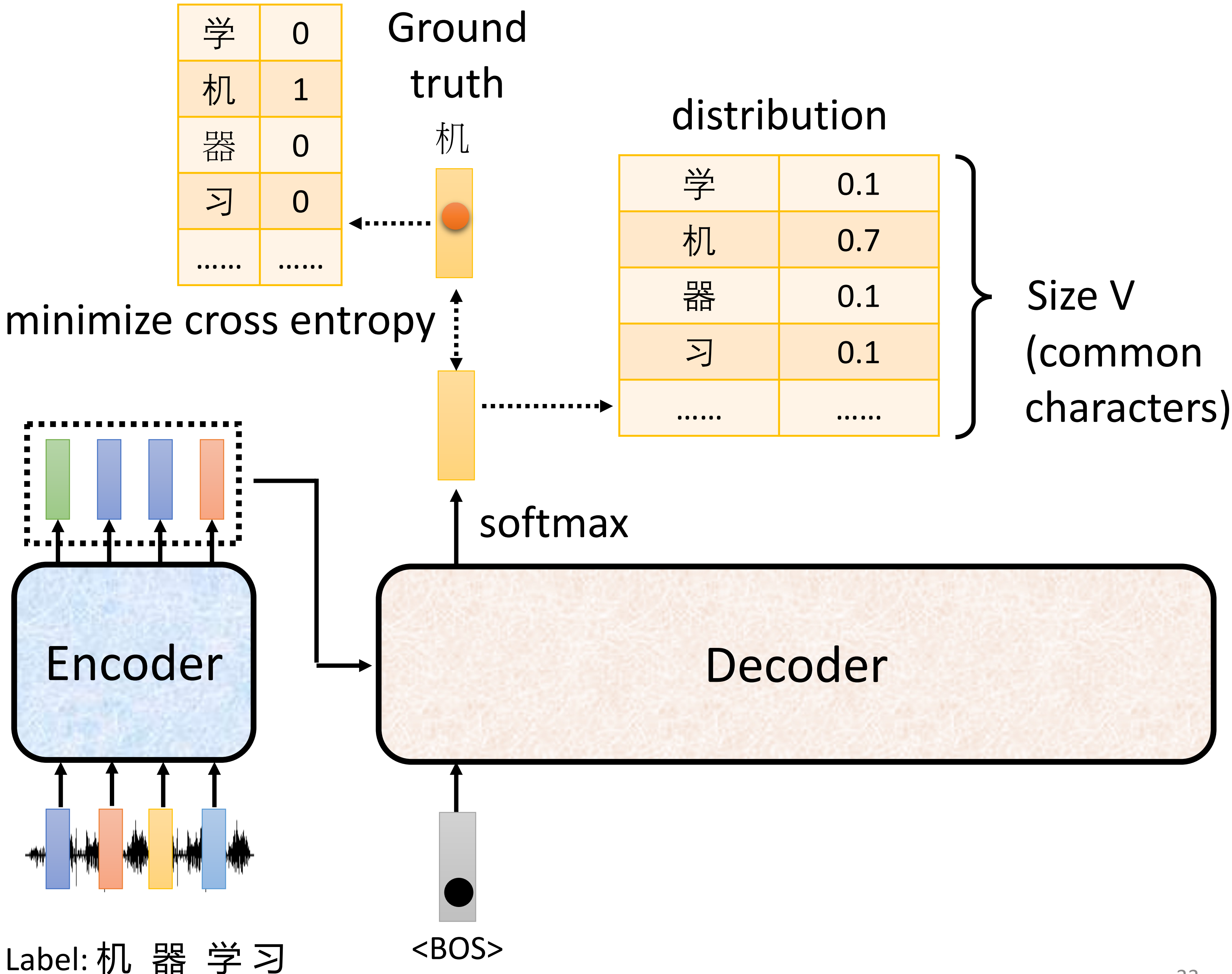
Training



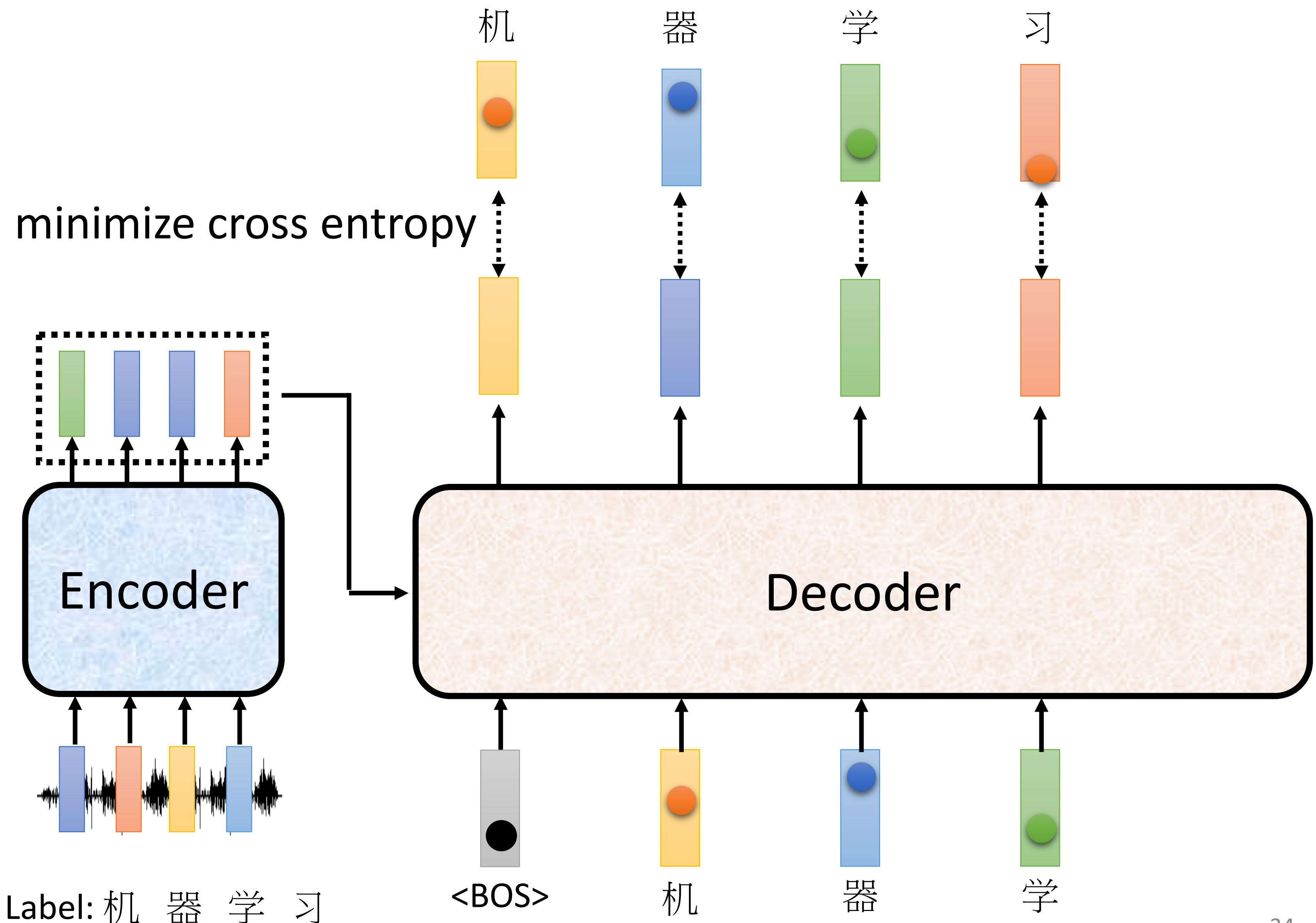
Training

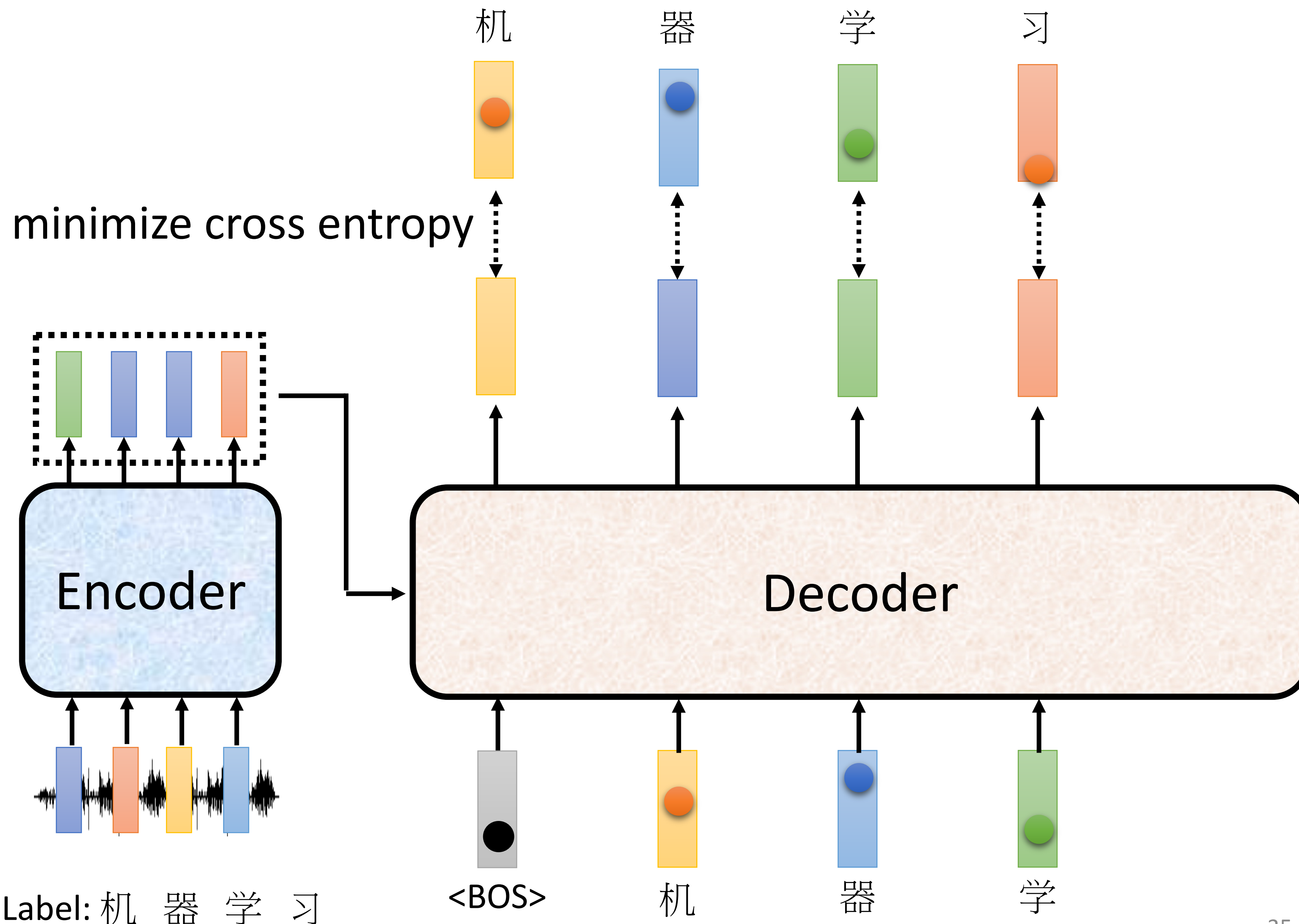


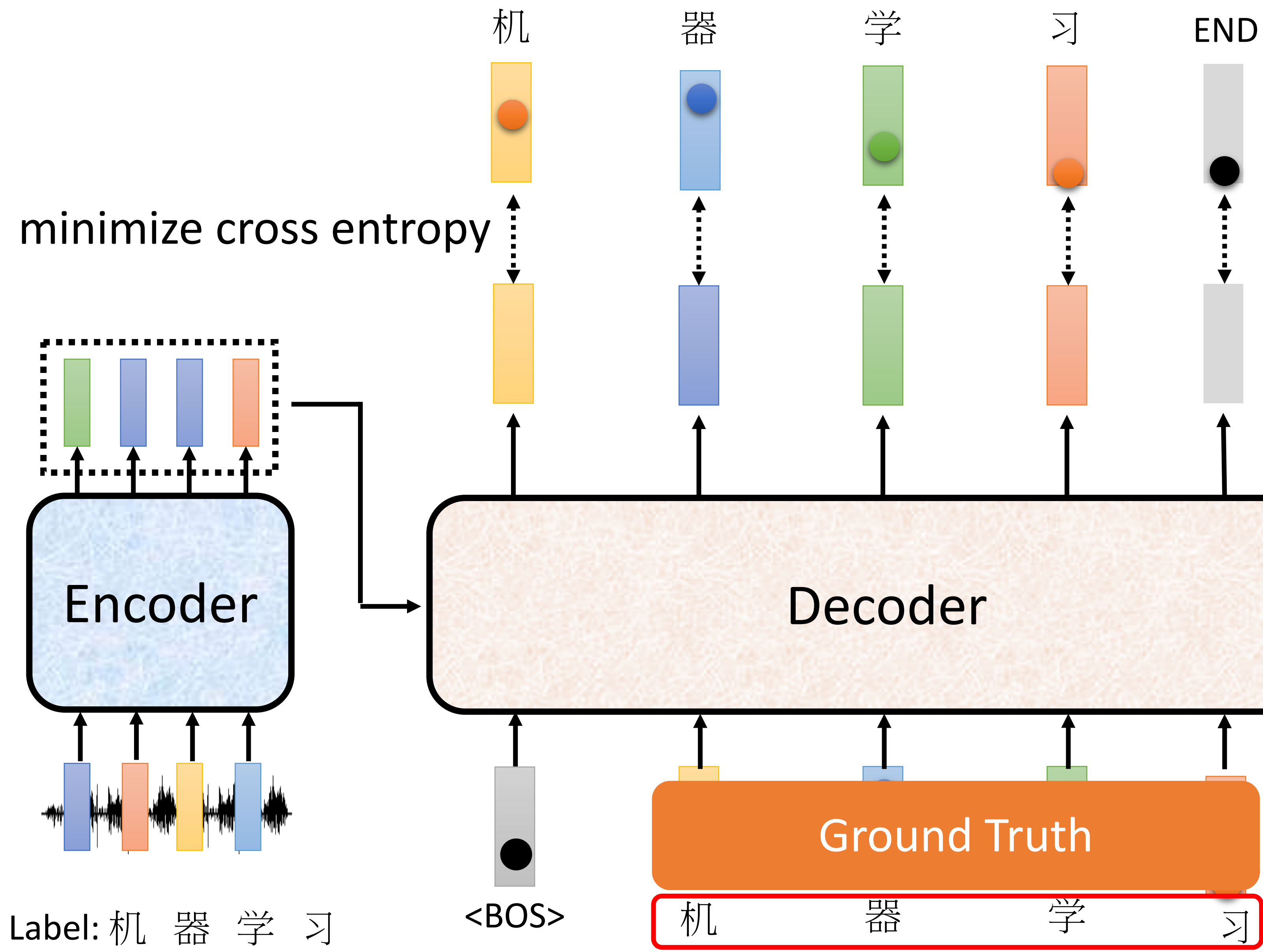
distribution
学 0.1
机 0.7
器 0.1
习 0.1
…… ……
Size V
(common
characters)
softmax
Encoder
Decoder
Label: 机 器 学 习
<BOS>



学 0
机 1
器 0
习 0
…… ……
Ground truth
机
minimize cross entropy
distribution
学 0.1
机 0.7
器 0.1
习 0.1
…… ……
Size V (common characters)
softmax
Encoder
Decoder
<BOS>
Label: 机 器 学 习


机
器
学
习
minimize cross entropy
Encoder
Decoder
<BOS>
机
器
学
Label: 机 器 学 习


机
器
学
习
minimize cross entropy
Encoder
Decoder
Label: 机 器 学 习
<BOS>
机
器
学



机
器
学
习
END
minimize cross entropy
Encoder
Decoder
Ground Truth
Label: 机 器 学 习
<BOS>
机 器 学 习

Thanks!

Q&A ?

wuxianyan@njau.edu.cn